

Technical Supplement I — The Classification Core

The Structure of Admissible Physics

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Companion to Paper 2: “The Structure of Admissible Physics”

Technical Supplement I of two; the foundational program is Technical Supplement II

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Abstract

This supplement is the classification core of Paper 2. Its citable result is a conditional classification theorem (Theorem 5.9) whose full input ledger is part of the statement: given the declared template caps and representation tables of the scan, the fixed gauge template $SU(3) \times SU(2) \times U(1)$, the imported generation count $N_{\text{gen}} = 3$, the imported one-loop Weyl beta coefficient, the anomaly and Witten equations as filter conditions, the F3 doublet–singlet content predicate, CPT identification, electromagnetic completeness (F6, an independent assumption in its canonical full-system non-degeneracy form), the Weyl-component objective, and the equal-unit channel convention, the Standard-Model five-multiplet template at 45 Weyl degrees of freedom is the unique minimum among the enumerated candidates, computed exactly. The abelian sector is closed by a-posteriori closure theorems for the selected template: given the scanned matter content, at least one abelian grading is necessary (under the named reading R-EC-inv with the entry-typed inventory I-typing), one suffices, the anomaly system fixes its direction up to normalization and relabeling, and no second factor is admissible under capacity minimality. The canonical F6 predicate is full-system non-degeneracy (10 survivors pre-CPT, 5 post-CPT); spectator reduction (8/4) and uniform dimension (4/2) are reported as robustness checks, with the same winner under all three. The capacity count $45 + 4 + 12 = 61$ is proved in the equal-unit channel convention; every cosmological reading of it is a bridge axiom (BA1–BA10, §3), and the cosmology-facing material lives at its corpus homes. Reproducibility is contractual: the scan is a finite enumeration in exact rational arithmetic, machine-verified in-tree and by a standalone script. The premise list itself — why these inputs and not others, and at what grade each is grounded — is the object of a separately presented program (Technical Supplement II (*The Foundational Gauge Program*)); this document consumes the inputs and proves the classification.

What Is Assumed, What Is Proved, What Is Deferred

Imported from Paper 1 Supplement. The four structurally necessary components of PLEC — A1 (finite capacity), MD (positive cost floor), A2 (argmin selection), BW (non-degeneracy) — together with the constitutive admissibility semantics (SP, K3, FD1–FD6) and the finite-carrier chain (T_{form} , T_{sep} , L_{ε^*} , L_{Δ} , T_{alg} , T_{GNS} , T2a–T2c, L_{loc} , L_{ω} , T_{adj} , L_{irr}). Self-contained re-proofs of the imported results live in the appendix of Technical Supplement II (*The Foundational Gauge Program*).

Consumed from Technical Supplement II (*The Foundational Gauge Program*) (proved there at the grades stated in its grade ledger).

1. L_{Cauchy} : cost-function uniqueness, $F(n) = n$; the channel unit for every cost comparison here.
2. The finite semisimple admissibility algebra and the gauge identification: every continu-

ous physical gauge redundancy is inner ($\text{Gau}_0 \subseteq \text{Inn}(\mathcal{A})$; equality under the named full-invariance hypothesis).

3. The exact sequence $1 \rightarrow Z(\text{U}(\mathcal{A})) \rightarrow \text{U}(\mathcal{A}) \rightarrow \text{Inn}(\mathcal{A}) \rightarrow 1$ and the relative-phase torus T_{rel} as the sole abelian source.
4. The matter category on the lifted group $\tilde{G}$, with the charge representation a named input (I-typing), and the lift and global $\mathbb{Z}_6$ quotient.
5. EC at $[\text{P}_{\text{structural_reading}} \mid \text{R-EC-inv} + \text{I-typing}]$ and CM at $[\text{P}]$ (A2 over the A1+MD admissible set).
6. $C(G) = \dim(G)$ under the operational cost premise pair (operational encoding; disjoint-anchor realizability).
7. Theorem R’s carrier requirements R1–R3, T_{7B} , $L_{\text{irr,uniform}}$, and $B1'$.

Core results proved in this supplement.

1. $L_{\text{gauge,unique}}$: gauge template uniqueness within the declared carrier filters (§4).
2. T_{field} : fermion content by complete 1,680-template scan of the declared space with seven filters (§5).
3. Theorem 5.9: the conditional classification theorem with its full input ledger (C1–C10) — the citable core.
4. The one- $\text{U}(1)$ a-posteriori closure theorems (§7).
5. $L_{\text{hypercharge}}$: hypercharge uniqueness from the anomaly system (§6).
6. L_{count} : capacity counting, $45 + 4 + 12 = 61$ (§10).
7. The asymptotic-freedom imported-coefficient check (§2).

Exited at the v4.0 core split (bridge material). Lorentzian signature, $d = 4$, Einstein equations, horizon equipartition, and saturation independence live at Papers 6, 8, and 41; this supplement retains the exit map and the bridge-axiom ledger (§3). The cosmology tail — the (61, 102) architectural reading, the sensitivity table, and the Planck comparisons — is staged for Papers 8 and 41 (exit note in §10).

Not proved here (honest flags).

1. Quantitative β -function derivation: only the imported coefficient check is here (§2); the capacity-level coefficient derivation is deferred to Paper 4A.
2. Cap completeness: the enumeration caps of the scan are declared inputs; that no admissible sub-45-DOF template exists outside them is an open proof obligation (§5.2).
3. Cosmological identifications: every cosmological reading of the 61 is a bridge axiom (BA5–BA7), not a result of this document.
4. Mass hierarchy, CKM/PMNS mixing, CP violation: Papers 4A/4B.
5. Confinement scale Λ_{QCD} and hadronization: Paper 4A.

Imported-ingredient hypothesis ledger (verified vs. assumed in the finite-cost environment)

Every off-framework mathematical import, with its hypothesis status *in this document’s finite setting*. “Verified” = the theorem’s hypotheses are checked here against the finite structures it is applied to; “assumed” = the hypothesis is taken as a premise and flagged where used.

Import	Applied to	Hypothesis status
Cartan closed-subgroup	classification class	verified (the class is $\prod_k \text{PU}(n_k)$ and closed subgroups; scope rider travels with <code>check_L_cost_gauge</code>)
Compact simple Lie classification	the F1–F3 table	imported whole (Killing/Cartan); carrier list extended to spinors in v4.0
One-loop β coefficient	F1/F2 filters	<i>imported, assumed</i> : structural sign pressure only is derived (§2); scan runs at the Weyl 2/3 convention
$N_{\text{gen}} = 3$	filter multiplicity	<i>imported, assumed here</i> : banked capacity–beta identities (§8); F1/F2 conditional on it
Electromagnetic completeness	F6 (canonical: full-system non-degeneracy)	<i>independent assumption</i> , declared at point of use (§5.3); logically independent of EC — computed
Compact-U(1) representation data	hypercharge normalization	<i>premise</i> : the abelian factor is compact with a chosen character lattice; the anomaly system fixes charge <i>ratios</i> , and the absolute unit ($Y_Q = 1/6$) is this convention (§6)
HKM / Malament / Kolmogorov / Lovelock	exited bridge	assumed as bridge axioms BA1–BA4 (§3); not consumed by the core
F2 rank window ($3 \leq k \leq \text{rank}(G)+1$)	the (F2) carrier filter	<i>selected filter premise</i> (F2-window): a declared physical filter — low-rank antisymmetric primitive invariants, few-body composite realizability (§4.2)
Operational cost premise pair	$C(G) = \dim(G)$ (cost ranking)	<i>named operational premises</i> : operational encoding (lower bound) + disjoint-anchor realizability (upper bound); Proposition 4.1

Terminology: four words with one fixed meaning each

faithful of a representation or state: trivial kernel. For states: $\omega(a^*a) = 0 \Rightarrow a = 0$ (GNS null ideal zero).

Never used to mean “physically realized.”

irreducible of a representation: no proper nonzero invariant subspace. Used only rep-theoretically; “primitive” is used for invariant tensors, never “irreducible.”

effective of a group action: only the identity acts trivially (synonym of faithful for actions); used when the group, rather than its representation, is the subject.

physically distinct (multiplet types) distinct isomorphism classes of $\tilde{G}$ -representations (Definition 2.20 of Technical Supplement II (*The Foundational Gauge Program*)); copies within a class (generations) are the same type at different multiplicity.

Reader’s map: one paper, two supplements

Paper 2’s technical material lives in two supplements. This document — Technical Supplement I, *The Classification Core* — carries finished mathematics on declared inputs: the conditional classification theorem with its full premise ledger, the scan and its waterfall, anomaly and hypercharge closure, the one-U(1) a-posteriori closure theorems, the Lie classification within the declared carrier filters, the capacity count, and the bridge-axiom ledger. Nothing in this document claims a derivation from A1/MD/A2/BW. Technical

Supplement II (*The Foundational Gauge Program*) carries the foundational program — the chain from the APF axioms toward this document’s inputs, each link at its banked grade, with its grade ledger and open-lane register. A reader auditing the classification needs this document and the premise ledger; a reader auditing the premises’ grounds needs Technical Supplement II.

Framing convention: “forces” and “selects” denote uniqueness, not processes

The supplement’s theorems and proofs use the verbs “forces,” “selects,” and “rejects” as standard mathematical shorthand for uniqueness results, arg min properties, and inadmissibility results respectively. Under the APF ontology, these verbs do not denote processes: nothing is forced by an agent, nothing is selected by a mechanism, nothing is rejected by a filter. Each theorem asserts a property of the set of admissible configurations—that it has a unique member, a minimum member, or no member satisfying some condition. For example, “capacity minimality selects $N_c = 3$ ” is shorthand for “ $N_c = 3$ is the arg min of $\dim(G)$ over R1–R3-admissible color groups,” and the alternative values ($SU(4)$, $SU(5)$, E_6 , ...) are *admissible* in the sense that each has cost below the capacity bound, but have higher cost than $SU(3)$ and are therefore not what is. The companion note *What Is, Is Minimum-Cost Admissible* spells this out in full. Readers may substitute the descriptive reading throughout the supplement without any change to the proofs.

Status of A2: argmin realization is a lawlike selection rule, not a derived dynamics

A2 is intentionally not derived in this supplement from a deeper dynamical variational principle. It is one of the four structural inputs of the finite-regime APF package: among admissible continuations satisfying A1, MD, and BW, the realized carrier is the minimum admissibility-cost representative. All selection claims in the supplement have the conditional form

$A1+MD+BW+A2+\text{listed interface hypotheses} \implies \text{unique minimum-cost admissible structure.}$

Thus the paper does not claim to prove that nature *must* obey global argmin realization from conventional QFT assumptions. It claims that, if A2 is accepted as the APF lawlike selection rule, the Standard-Model template is the unique admissible minimum under the stated finite ledger. This is why counterfactuals that change A2, make new generators cost-free, or subtract structural gauge generators from the ledger are not small variants of the theorem; they define a different selection problem.

Self-contained Standard Model datum. Section 6 derives the unique anomaly-free hypercharge assignments for one generation of the Standard Model fermion template purely from the tools assembled in this supplement (gauge template from §4, fermion content from §5, anomaly conditions from §6). The result ($Y_Q = 1/6$, $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$) constitutes a nontrivial, experimentally verified prediction of the derivation chain: the charge *ratios* and their quantization are derived, while the absolute unit is fixed by the compact- $U(1)$ charge-lattice convention (front-matter imported-ingredient ledger).

Dependency table. Each row gives a result proved in this supplement, its inputs, and the downstream results that depend on it.

Theorem	Inputs	Used by
$L_{\text{gauge,unique}}$ (§4)	Theorem R, L_{Cauchy} , Lie classif. (all consumed from Technical Supplement II (<i>The Foundational Gauge Program</i>) / imported)	T_{field} , L_{count}
T_{field} (§5)	C1–C10 ledger (Thm. 5.9)	closure theorems, L_{count}
One-U(1) closure (§7)	T_{field} output; EC/CM at their grades	abelian-sector consistency
$L_{\text{hypercharge}}$ (§6)	T_{field} , anomaly conditions	closure theorems, L_{count}
L_{count} (§10)	T_{field} , T_{Higgs} , $L_{\text{gauge,unique}}$, BA5 convention	I2; exited bridge readings

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On operational language. Operational verbs are local readings of an underlying admissibility-space structure under a temporal slicing. They are retained because they are the working vocabulary of physics; the structural reading is always available: “acts,” “forms,” “selects,” and “evolves” mean that the corresponding admissible configuration, quotient, minimum, or continuation relation exists. This convention is inherited from Paper 0 and Paper 1 Supplement v8.24.

Part I

Classification Inputs and Conditional Bridges

1. Consumed inputs: the foundational chain at a glance

This document proves the classification and closure results and consumes the foundational chain as cited premises. The chain itself — statements, proofs, grades, and named premises — is Technical Supplement II (*The Foundational Gauge Program*); nothing here claims a derivation from the APF axioms A1/MD/A2/BW. The box below states what is consumed and at what grade.

Chain outputs consumed by this document

1. **The cost unit.** L_{Cauchy} : $F(n) = n$ on the discrete channel domain; every cost comparison in this document is a channel count. Grade [P].
2. **The algebra and the gauge identification.** A finite semisimple admissibility algebra $\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$ (standard C^* route, given the IJC occupancy, which is empirical), with every continuous physical gauge redundancy inner: $\text{Gau}_0 \subseteq \text{Inn}(\mathcal{A})$, and equality under the named full-invariance hypothesis. The SM specialization $\{n_k\} = \{3, 2, 1\}$ enters this document as hypothesis H1 of the closure theorems, not as a theorem of this document.
3. **The exact sequence and the abelian source.** $1 \rightarrow Z(\text{U}(\mathcal{A})) \rightarrow \text{U}(\mathcal{A}) \rightarrow \text{Inn}(\mathcal{A}) \rightarrow 1$; continuous abelian gauge freedom lives in the relative-phase torus T_{rel} , physical exactly where charged matter sees it.
4. **EC and CM at their grades.** Enforcement completeness at $[\text{P}_{\text{structural_reading}} \mid \text{R-EC-inv} + \text{I-typing}]^a$; capacity minimality at [P] (A2 over the A1+MD admissible set, stated at channel-configuration level).
5. **The gauge-cost equality.** $C(G) = \dim(G)$ under the operational cost premise pair — operational encoding for the lower bound, disjoint-anchor realizability for the upper bound — realized in Proposition 4.1^b.
6. **Carrier requirements and the matter category.** Theorem R's carriers (complex of dimension ≥ 3 with a low-rank antisymmetric primitive invariant; pseudoreal of dimension 2; a single abelian grading) and the matter category on the lifted group $\tilde{G}$, with the charge representation a named input (I-typing). The Sep/IJC branch material is consumed the same way: the bridge from superadditivity to a noncommutative algebra is derived given the IJC occupancy, and the occupancy is empirical — read off the record, not derived from A1. The grade-ledger table near the front of Technical Supplement II (*The Foundational Gauge Program*) is the one-page map of all of the above.

^aCode reference: `check_L_EC_inventory_reading` in `ec_inventory_reading.py`.

^bCode reference: `check_L_cost_gauge` in `core.py`.

2. Asymptotic freedom: structural sign heuristic and imported coefficient check

Paper 2 §3.6–3.7 derives the structural necessity of asymptotic freedom from the non-abelian self-interaction of admissibility channels: gauge bosons carry charge, producing anti-screening that overwhelms matter screening. The full quantitative derivation (one-loop β -coefficient from Casimir arithmetic) is in Paper 4A ($L_{\text{beta},\text{capacity}}$). This section provides the schematic sign calculation using only the matter content derived in this paper.

Proof boundary. The one-loop β -function formula is a structural result of gauge theory. Paper 4A derives it from capacity arguments ($L_{\text{beta},\text{capacity}}$); here we use it as an imported formula (verifiable in any QFT textbook) and evaluate it with the derived matter content.

The one-loop β -function coefficient for a gauge group factor G_i with adjoint Casimir C_A is stated once, in the Weyl convention used everywhere in this document (the scan’s F1–F2 filters included):

$$b_0 = \frac{11}{3} C_A - \frac{2}{3} \sum_r T(r), \quad (1)$$

where the sum runs over the left-handed Weyl fermion species charged under G_i and $T(r)$ is the Dynkin index of the representation each carries. Complex scalars contribute a separate term $-\frac{1}{3} \sum_s T(s)$; we keep it out of the fermion displays below and account for it once, at the end. The textbook Dirac form $b_0 = (11/3)C_A - (4/3)T_R N_f$, with N_f counting Dirac flavors, is the same number repackaged (one Dirac flavor is two Weyl species); mixing the two conventions — the Dirac coefficient on a Weyl count — is exactly the mispricing the F1 filter statement warns against (§5.3). Asymptotic freedom holds when $b_0 > 0$: the gauge self-interaction (the C_A term, from non-abelian structure) overwhelms the matter screening (the $\sum T$ term).

Proposition 2.1 (Sign of b_0 for the derived matter content). *With the SM fermion content derived in T_{field} (§5), both non-abelian factors are asymptotically free:*

$$b_0^{\text{SU}(3)} = \frac{11}{3} \cdot 3 - \frac{2}{3} \cdot 6 = 11 - 4 = 7 > 0, \quad (2)$$

$$b_0^{\text{SU}(2)} = \frac{11}{3} \cdot 2 - \frac{2}{3} \cdot 6 = \frac{22}{3} - 4 = \frac{10}{3} > 0, \quad (3)$$

with $\sum_r T(r) = 6$ for each factor, computed in the proof.

Proof. For $\text{SU}(3)$: $C_A = 3$. Each generation carries four colored Weyl species — the two $\text{SU}(2)$ components of Q (each a color fundamental) plus u^c and d^c — so $\sum_r T(r)$ per generation is $4 \times \frac{1}{2} = 2$, and over three generations $\sum_r T(r) = 6$. Therefore $b_0^{\text{SU}(3)} = 11 - (2/3)(6) = 11 - 4 = 7$.

For $\text{SU}(2)$: $C_A = 2$. Each generation carries four Weyl doublets — Q in three colors plus L — so $\sum_r T(r)$ per generation is $4 \times \frac{1}{2} = 2$, and over three generations $\sum_r T(r) = 6$: the matter contribution is $(2/3)(1/2)(12) = 4$. Therefore $b_0^{\text{SU}(2)} = 22/3 - 4 = 10/3$.

The scalar term is separate and does not change either sign: the single complex Higgs doublet contributes $-\frac{1}{3} \cdot \frac{1}{2} = -\frac{1}{6}$ to $b_0^{\text{SU}(2)}$ (giving $19/6 > 0$ with the fermions) and nothing to $b_0^{\text{SU}(3)}$.

Both coefficients are positive. The signs depend only on the matter content (derived in T_{field}) and the group theory (Casimirs of $\text{SU}(3)$ and $\text{SU}(2)$), not on any free parameter. $\square$

What this establishes and what it does not. The sign of b_0 determines the qualitative running: positive b_0 means the coupling decreases at high energy (asymptotic freedom) and grows at low energy (confinement regime for SU(3)). This section does *not* derive the formula (1) from capacity arguments — that derivation is in Paper 4A ($L_{\text{beta},\text{capacity}}$). Here the formula is used as an imported result (standard in any QFT textbook [6]) and evaluated with the derived matter content. The structural claim of Paper 2 — that non-abelian self-interaction generically produces asymptotic freedom when the matter content is sufficiently small — is confirmed by this explicit calculation.¹

2.1 Comparison with established β -function derivations

The one-loop β -function coefficient $b_0 = (11/3)C_A - (4/3)T_R N_f$ has been derived by at least four independent methods in the literature. This subsection identifies each, states what the APF reuses vs. reinterprets, and clarifies the boundary between established results and the APF’s contribution.

1. Feynman-diagram computation (Gross–Wilczek, Politzer, 1973). The original derivation computes the one-loop vacuum polarization of the gluon propagator via three diagrams: the fermion loop (screening, $-4/3 T_R N_f$), the ghost loop (antiscreening, $+1/3 C_A$), and the gluon self-energy with a three-gluon vertex (antiscreening, $+10/3 C_A$). The sum gives $b_0 = (11/3)C_A - (4/3)T_R N_f$.

What the APF reuses: the numerical result $b_0 = (11/3)C_A - (4/3)T_R N_f$. This is imported as an established formula (Proposition 2.1) and evaluated with the derived matter content. The APF does not re-derive this formula in the present paper.

What the APF reinterprets: the physical mechanism. In the standard derivation, the 11/3 arises from the sum of gluon and ghost loop contributions; the ghost loop is a gauge-fixing artifact with no direct physical interpretation. In the APF’s capacity language (§4.1 of Technical Supplement II (*The Foundational Gauge Program*)), the antiscreening is attributed to the *superadditive cost of co-located non-abelian channels* ($L_\Delta > 0$), which is a physical statement about admissibility economics, not a gauge-fixing artifact. The ghost contribution is absorbed into the capacity self-cost.

What is genuinely new: the sign of b_0 is evaluated with no free choice left in the matter input. In standard QCD, the sign of b_0 depends on the number of quark flavors N_f , which is a free input. Here N_f is fixed within the ledger (T_{field} : 6 flavors from the scan, under the C1–C10 premises), and the sign follows by evaluating the imported one-loop formula with those ledger-fixed inputs — sign pressure with imported weights, exactly the banked check’s register. The novelty is not in the formula but in the fact that its inputs are the ledger’s, not free choices.

2. Background-field method (Abbott, 1981; ’t Hooft, 1976). The background-field method computes the β -function by expanding around a classical background $\bar{A}_\mu$ and integrating out quantum fluctuations. The result is gauge-invariant at each step, making the antiscreening origin more transparent: the three polarization states of a massive spin-1 particle contribute $+(11/3)C_A$ (two physical transverse polarizations contribute $+2C_A$, and the longitudinal / temporal sector contributes $+(5/3)C_A$ after ghost subtraction).

APF relationship: the background-field decomposition has a direct capacity analogue. The “background field” corresponds to the admissibility structure at a fixed interface (the cost kernel g); the “quantum fluctuations” are perturbations of the distinction structure. The $(11/3)C_A$ counts the capacity cost of maintaining p self-interacting channels in $d = 4$ spacetime dimensions: two physical

¹Code reference: `check_T_confinement` in `gauge.py`.

polarizations (from the transversality condition, which is the admissibility analogue of gauge fixing) plus the longitudinal capacity overhead. Paper 4A makes this correspondence precise.

3. Index-theoretic derivation (Atiyah–Singer, heat-kernel methods). The β -function can be extracted from the heat-kernel expansion of the functional determinant $\det(-D^2)$, where D is the gauge covariant derivative. The Seeley–DeWitt coefficient $a_2(D^2)$ gives the one-loop divergence, and its group-theoretic content reproduces b_0 . This approach makes no reference to individual Feynman diagrams; the result follows from the spectral geometry of the gauge-covariant Laplacian.

APF relationship: the index-theoretic derivation is the closest mathematical relative of the APF’s capacity approach. Both compute a spectral invariant of the gauge connection: the Seeley–DeWitt coefficient is a spectral invariant of $-D^2$, while the APF’s capacity cost is a spectral invariant of the admissibility algebra’s cost kernel B (via the eigenvalue decomposition in Lemma 2.22 of Technical Supplement II (*The Foundational Gauge Program*)). The capacity inequality (eq. 6 of Technical Supplement II (*The Foundational Gauge Program*)), $p \cdot C_A > T_R \cdot N_f^{\text{total}}$, is a statement about the spectral content of the gauge sector vs. the matter sector, analogous to the comparison of a_2 contributions in the heat-kernel expansion. The precise coefficient $11/3$ corresponds to the a_2 coefficient of the spin-1 Laplacian in $d = 4$, which Paper 4A derives from the polarization count (the same $d = 4$ that the APF derives in the exited $d=4$ argument, Paper 6; §3).

4. Twistor-space methods (Penrose, 1967; Witten, 2004). Witten’s reformulation of perturbative gauge theory in twistor space computes scattering amplitudes without Feynman diagrams, using holomorphic curves in $\mathbb{CP}^3$. The one-loop β -function can be extracted from the holomorphic anomaly of the twistor string. The result agrees with the standard computation but organizes the contributions differently: the antiscreening is attributed to the self-dual sector of the gauge field, and the screening to the matter multiplets’ contribution to the holomorphic anomaly.

APF relationship: the twistor approach is structurally different from the APF (twistor space is infinite-dimensional and uses complex geometry; the APF works with finite-dimensional admissibility algebras). However, both share a feature: they derive the β -function sign from *structural* properties of the gauge field (self-duality in the twistor case, self-interaction cost in the APF case) rather than from individual diagram counting. The APF does not use twistor methods; the comparison is noted to show that the impulse to derive b_0 from structural principles rather than diagram-by-diagram computation has precedent.

Summary: new vs. reinterpreted.

Element	Status	Detail
Formula $b_0 = (11/3)C_A - (4/3)T_R N_f$	Imported	Standard one-loop result (Gross–Wilczek–Politzer, 1973); not rederived here.
Sign $b_0 > 0$ for SM content	Imported + derived inputs	Formula imported; matter content N_f derived from T_{field} . The sign is therefore a <i>prediction</i> (given A1), not a parameter choice.
Capacity inequality $pC_A > T_R N_f^{\text{total}}$	New	Derived from L_Δ (superadditivity) and Lemma 2.22 of Technical Supplement II (<i>The Foundational Gauge Program</i>) (generator–channel bridge). Provides a sufficient condition for $b_0 > 0$ without the formula.
Antiscreening from L_Δ	New interpretation	The superadditive cost of co-located non-abelian channels provides a physical mechanism for the C_A term; reinterprets the ghost + gluon-loop sum as admissibility self-cost.
Precise coefficients 11/3, 4/3	Deferred	Paper 4A ($L_{\text{beta, capacity}}$) derives these from polarization counting in $d = 4$ and the spin-statistics connection. Not available in this supplement.
Index-theoretic parallel	Noted	The capacity cost is a spectral invariant of B , analogous to the Seeley–DeWitt a_2 coefficient. Not developed further here.

The honest assessment is: the APF’s contribution to the β -function story in this supplement is *modest*. The formula is imported; the sign is verified with derived inputs; the capacity inequality provides a new sufficient condition but does not reproduce the precise coefficients. The genuine novelty is that N_f (and hence the sign of b_0) is derived rather than assumed. The reinterpretation of antiscreening as admissibility self-cost is a new physical picture but does not yet produce quantitative predictions beyond what the standard formula gives. Paper 4A is where the capacity-based derivation must stand or fall.

Paper 4A coefficient program. The deferred coefficient derivation has three explicit steps: (i) construct the capacity-level analogue of a Wilsonian shell by resolving which continuation channels become active under refinement; (ii) prove that the adjoint self-correlation contribution is weighted by the same channel-count invariant as C_A , while a matter channel is weighted by the finite-index embedding invariant corresponding to T_R ; and (iii) show regulator independence by proving that refinements with the same representation index produce the same shell increment. Already testable intermediate results are therefore the signs, the C_A/T_R weighting pattern, and the fact that adding enough charged matter flips the sign exactly as the standard coefficient predicts. This supplement imports the coefficient but records these as the Paper 4A proof obligations.

3. Conditional bridges: the exit map and bridge axioms

This supplement’s claim surface ends with the gauge/matter template and its capacity accounting. Everything downstream of that surface — causal order, continuum spacetime, Lorentzian signature, Einstein dynamics, horizon thermodynamics, cosmological density fractions, and the structural predictions (vacuum stability, proton stability, strong CP) — is a *bridge*: a conditional extension whose hypotheses are not theorems of this document. Earlier versions carried the bridge derivations in full; they now live with their corpus homes (Papers 6, 8, 41, 45, and 46), and what remains here is the boundary statement and the explicit list of bridge axioms, each with its named import and its failure mode.

Theorem 3.1 (Conditional spacetime bridge, not an input to gauge recovery). *The finite gauge/-matter results of this supplement do not depend on the Lorentzian, $d = 4$, Einstein-equation, horizon, or cosmological-density claims. Those claims form a downstream bridge: if the positive cost kernel, irreversible ordering, locality, and the imported HKM/Malament/Lovelock hypotheses are all satisfied, then the cost-geometry interpretation follows. If any bridge hypothesis fails, the gauge/matter derivation remains intact and only the spacetime/cosmology interpretation is downgraded.*

Why this is one-way. The gauge/matter derivation uses only the finite admissibility ledger, Sep/IJC routing, the finite $\ast$ -algebra carrier, inner automorphisms, the classification of compact Lie algebras, and the finite template scan. None of those steps invokes a continuum manifold, Lorentzian signature, Lovelock dynamics, or horizon thermodynamics. The bridge consumes the derived cost kernel and irreversibility order as inputs; nothing flows back.

3.1 The bridge-axiom ledger

Each axiom below is a premise the bridge *assumes*. None is derived from A1, MD, A2, or BW in this document. A reader who rejects a bridge axiom loses exactly the row’s conclusion and nothing upstream.

BA1 Causal order. The commitment relation $\prec$ is irreflexive and transitive. *Import:* order axioms — irreversible positive commitment alone does not prove irreflexivity (applying the identity is not an undoing operation), and transitivity needs a composition-monotonicity premise. *Failure mode:* $\prec$ admits cycles or fails to compose; no causal set. *Home:* Paper 6.

BA2 Continuum structure. The event set carries a C^2 differentiable manifold structure compatible with the projective-limit measure. *Import:* the manifold itself — Kolmogorov extension yields a measure on a product-limit space, not charts; tangent spaces, exponential maps, and C^2 regularity are assumed with it. *Failure mode:* the continuum description has no differentiable refinement; the metric chain stops at the order. *Home:* Paper 6.

- BA3 Lorentzian reconstruction hypotheses.** $(M, \prec)$ satisfies the Hawking–King–McCarthy/Malament distinguishing conditions. *Import:* HKM 1976 / Malament 1977 hypotheses. *Failure mode:* the order is not realizable as the causal order of any smooth Lorentzian manifold. *Home:* Paper 6.
- BA4 Second-order dynamics.** Locality, diffeomorphism covariance, conservation, and second-order metric dependence — the Lovelock premises. *Import:* Lovelock 1971 hypotheses; a finite information bound does not by itself limit a continuum response law to second derivatives. *Failure mode:* higher-derivative or nonlocal gravitational dynamics. *Home:* Paper 6.
- BA5 Common capacity unit.** Each Weyl fermion component, each real Higgs component, and each gauge generator occupies exactly one capacity slot of equal weight, so that $45+4+12 = 61$ is a count in a single currency. *Import:* the unit-equivalence premise. *Failure mode:* sector-dependent slot weights; the 61 loses its privileged status. *Home:* Paper 8.
- BA6 Cosmological identification.** Capacity-slot fractions are Friedmann density parameters ($\Omega_\Lambda = 42/61$ and kin). *Import:* a phenomenological identification (a reading, not a theorem); maximizing entropy over 61 equal slots gives a uniform distribution over *slots*, and the step from slot fractions to cosmological densities is an ansatz. *Failure mode:* the fractions are bookkeeping only. *Home:* Paper 8 (dark-sector reading: Paper 35).
- BA7 Horizon equipartition conditions.** The maximum-entropy equal-priors conditions under which the horizon ledger equidistributes. *Import:* the equipartition assumption set of the exited horizon section. *Failure mode:* weighted priors; the equipartition fractions shift. *Home:* Paper 41.
- BA8 Saturation conditionality (baryon number).** The proton-stability conclusion is conditional on full Bekenstein saturation *and* on the completeness of the derived particle content: within that content no mediator generates the $\Delta B = 1$ dimension-six operators, but those operators are gauge-invariant under $SU(3) \times SU(2) \times U(1)$ as EFT structures, and ultraviolet degrees of freedom outside the derived content could source them. The honest claim is *no generation within the derived content*, not non-existence of the operators. *Home:* bank checks `check_T_proton`, `check_L_B_operator_exclusion` (`gauge.py`).
- BA9 Vacuum bridge.** The exited vacuum-stability argument shows uniqueness of the minimum of the APF potential under its own assumptions; it does not by itself constrain the renormalization-group-improved Standard Model effective potential. *Home:* Paper 45.
- BA10 Strong CP mechanism.** $\theta_{\text{QCD}} = 0$ is a proposed mechanism carried at [P-heuristic]; the admissibility principles stated here do not derive the absence or irrelevance of the θ -term. *Home:* Paper 46.

3.2 Exit map

Territory	Was (v3.9)	Home	Bridge axioms
Causal order $\rightarrow$ Lorentzian $\rightarrow d=4 \rightarrow$ Einstein	§9	Paper 6 + Supp.	BA1–BA4
MECE partition ($61 = 42 + 3 + 16$)	§17	Paper 8	BA5
Cosmological fractions ($\Omega_\Lambda = 42/61$)	§17	Paper 8 / 35	BA5, BA6
Horizon equipartition (L_{equip})	§18	Paper 41	BA7
Saturation independence ($L_{\text{sat,part}}$)	§19	Paper 8	BA5
Vacuum stability	§20	Paper 45	BA9
Proton stability	§20	bank (<code>gauge.py</code>)	BA8
Strong CP	§20	Paper 46	BA10

The bank checks for the exited results (`check_Delta_signature`, `check_T8`, `check_T9_grav`, `check_P_exhaust`, `check_L_equip`, `check_L_saturation_partition`, `check_T_vacuum_stability`, `check_T_theta_QCD`, `check_T_proton`) remain banked and machine-verified; this supplement simply no longer presents their derivations as its own claim surface.

Part II

Gauge Template and Matter Content

4. Gauge template uniqueness: Lie classification within the declared carrier filters

Paper 2 §5.1 states $L_{\text{gauge,unique}}$ in compact form. This section provides the full classification with explicit exclusion reasons across the four infinite families and five exceptional algebras (the bank check tests a 17-representative list; the families are handled by closed-form monotonicity), quantified capacity costs, and closed-form proofs for every elimination.

The capacity cost of a gauge group G is $C(G) = \dim(G) \cdot F(1)$ where F is the unique cost function from L_{Cauchy} (Theorem 1.3 of Technical Supplement II (*The Foundational Gauge Program*)). Since $F(1) = 1$ (unit normalization), $C(G) = \dim(G)$ — an equality resting on the two named operational premises (operational encoding for the lower bound; disjoint-anchor realizability for the upper bound), composed in Proposition 4.1. This is the quantity minimized in the classification below.

Proposition 4.1 (Gauge generators are separated admissibility interfaces). *Under the operational-encoding and disjoint-anchor-realizability premises (the operational cost premise pair of the front-matter ledger), for any compact Lie group G with Lie algebra $\mathfrak{g}$ of dimension n , the n independent generators $\{T^a\}_{a=1}^n$ constitute n separated admissibility interfaces in the sense of L_{Cauchy} ’s additivity premise (C1): for any pair $a \neq b$, committing admissibility to T^a does not reshape the channel landscape seen by T^b , so the superadditive surplus $\Delta(T^a, T^b) = 0$. Therefore the total cost is additive: $C(G) = \sum_{a=1}^n F(1) = \dim(G)$.*

Proof. The exact count is the banked gauge clause: $n(G) = \dim(G)$, hence $C(G) = \dim(G) \cdot \varepsilon$ in

$F(1) = 1$ units². The lower bound $C(G) \geq \dim(G)$ is Lemma 2.22 of Technical Supplement II (*The Foundational Gauge Program*) (resolution rank: invariance of domain on the $\dim(G)$ -dimensional group manifold — basis-independent, commutativity-independent), and it consumes that lemma’s named operational-encoding premise: a continuous injective operational encoding of the local group action is what invariance of domain applies to. The upper bound (no superadditive surplus *across generator channels beyond the count*) is the C1-additivity clause: what must be checked is that the $\dim(G)$ resolution channels are separated interfaces in L_{Cauchy} ’s sense, i.e. that committing one does not reshape the landscape of another ($\Delta = 0$). By T_M (interface monogamy) and L_{loc} , channels with disjoint anchor supports are separated. That the $\dim(G)$ resolution distinctions can be anchored on disjoint supports is a named *construction premise* (disjoint-anchor realizability), not a consequence of coordinate independence: the group coordinates are independent by construction, but independence of coordinates does not by itself deliver physical disjointness of anchor supports, so the realizability of a disjointly anchored channel family is stated as the premise the upper bound consumes. Earlier versions argued the separation from Killing-orthogonality of the generators; that argument is demoted — Killing orthogonality is a Lie-algebra inner-product statement, it is not by itself orthogonality of the admissibility projections under the anchor form B , and the Killing form is degenerate on abelian factors. The separation claim is carried by the disjoint-anchor realizability premise (and, at the bank level, by the two proof routes of `check_L_cost_gauge`, which carry the same premise structure: the check verifies the stated dimensions and arithmetic, not the encoding or realizability premises); Killing orthogonality is retained only as the natural bookkeeping basis. Therefore the cost is linear in $\dim(G)$ for every group in the classification class, including E_6 at 78. $\square$

Remark. This proposition closes an otherwise tacit step: without it, the cost formula $C(G) = \dim(G)$ would implicitly assume that gauge generators are C1-admissible without derivation. The capacity-minimality comparisons throughout this section (E_6 : 78 vs. $\text{SU}(3)$: 8, etc.) all rely on Proposition 4.1; it is the bridge from L_{Cauchy} ’s $\mathbb{N}$ -domain additivity to the linearity of cost in $\dim(G)$.

4.1 Setup: the classification problem

Theorem R (§3.5 of Technical Supplement II (*The Foundational Gauge Program*)) provides three abstract carrier requirements:

- R1** A faithful complex carrier of dimension ≥ 3 with irreducible totally antisymmetric invariant of rank $k \geq 3$.
- R2** A faithful pseudoreal carrier of dimension exactly 2.
- R3** A single abelian factor ($\text{U}(1)$) with distinct charge assignments for all physical multiplets.

R1–R3 are carried in from the foundational chain (Theorem R, §3.5 of Technical Supplement II (*The Foundational Gauge Program*)), where their derivations and grades live; this section consumes them as the classification’s carrier requirements. The problem is now purely mathematical: *which compact Lie group(s) can host all three carriers simultaneously, at minimum total capacity cost?*

The capacity cost of a gauge group G is $\dim(G) \cdot \varepsilon^*$ (L_{Cauchy} , Theorem 1.3 of Technical Supplement II (*The Foundational Gauge Program*)), under the operational cost premise pair of Proposition 4.1): the number of independent gauge generators, each costing one admissibility channel. This cost functional is unique (no free parameter) and is the quantity to be minimized.

²Code reference: `check_L_cost_gauge` in `core.py`.

4.2 Classification of the confining factor over the declared carriers (R1)

Input. The classification of compact simple Lie algebras is a completed theorem of pure mathematics. There are exactly four infinite families (A_n, B_n, C_n, D_n for $n \geq 1, 2, 3, 4$ respectively) and five exceptional algebras (G_2, F_4, E_6, E_7, E_8). For each, the fundamental (lowest-dimensional faithful) representation has a known dimension and reality type (real, pseudoreal, or complex). The scan below does *not* stop at fundamentals: for the orthogonal series the chiral spinor representations are faithful carriers of the spin groups and must be tested in their own right — type D_{2k+1} has *complex* chiral spinors (Spin(10)'s **16** is the physical case), type D_{2k} real or pseudoreal ones, and types B_n, C_n contribute no complex spinors. Earlier versions tested only the vector-type fundamentals of the D series; the spinor rows are now in the table. This data is imported from standard references [3, 4] and is not framework-dependent.

One boundary is stated once, honestly: no highest-weight exhaustiveness theorem over all faithful complex irreducible carriers is claimed. The examined carrier set is the fundamentals of the four families and the five exceptional algebras, plus the stated spinor rows of the orthogonal series; candidates outside that set are eliminated by the cost ranking (under the operational premise pair of Proposition 4.1), not by a representation-theoretic exclusion theorem.

Filters. R1 imposes three necessary conditions on the confining factor G_{conf} :

- (F1) The fundamental representation must be *complex* ($V \not\cong V^*$ as G -modules). This is forced by $B1'$ (Theorem 3.6 of Technical Supplement II (*The Foundational Gauge Program*)): real and pseudoreal carriers cannot sustain oriented composites.
- (F2) The representation must admit a totally antisymmetric *primitive* invariant of rank $k \geq 3$ — primitive meaning an invariant that supports a k -body neutral composite of the carrier at small k , exemplified by the rank- N ε -tensor of the $SU(N)$ fundamental. Stated precisely, once and for all: (F2) requires a totally antisymmetric *primitive* invariant in $V^{\otimes k}$ at rank $3 \leq k \leq \text{rank}(G) + 1$, where primitive means not a power or product of lower-rank invariants. For the $SU(N)$ fundamental the ε -tensor sits at $k = N = \text{rank}(G) + 1$ and is primitive; the top exterior power of a generic complex carrier sits at $k = \dim V \gg \text{rank}(G) + 1$ and fails the rank bound. Two precisions, both corrections to earlier versions: (i) the invariant must be totally *antisymmetric* — a symmetric invariant does not produce a fermionic composite of identical carriers (E_6 's cubic on the **27** is symmetric and does NOT satisfy this filter; the v3.9 table marked it as a rank-3 pass, which was wrong); (ii) a bare existence condition at unrestricted rank is vacuous for complex carriers, because the top exterior power $\Lambda^{\dim V} V$ of any representation of a connected semisimple group is invariant (trivial determinant character). The filter's selective content is therefore antisymmetry at *low* rank: for the $SU(N)$ fundamental the ε -invariant sits at $k = N$, a physical few-body composite for the small N the cost ranking keeps alive; candidates whose only antisymmetric invariant is a large-rank volume form gain nothing physical from it, and the cost ranking eliminates them independently. Bilinear ($k = 2$) invariants are excluded because they produce self-conjugate composites ($B = \bar{B}$). The rank window itself is a *selected filter premise* — named here **F2-window** and carried in the imported-ingredient ledger: requiring the primitive antisymmetric invariant to sit at $3 \leq k \leq \text{rank}(G) + 1$ is a declared physical filter (few-body neutral composites of the carrier must exist at low rank), motivated by the composite-realizability reading of R1, and it is a premise of the classification, not a consequence of Lie theory.
- (F3) The representation must be *faithful*: every non-identity group element acts non-trivially. This is forced by admissibility completeness at its grade (EC, $[P_{\text{structural_reading}} \mid \text{R-EC-inv} +$

I-typing]): if a gauge transformation acts trivially on all matter, it is physically undetectable and should not be counted.

Theorem 4.2 ($L_{\text{gauge,unique}}$ — Step 2: Confining Factor Classification). *The only compact simple Lie algebra whose fundamental representation satisfies (F1)–(F3) is $A_n = \mathfrak{su}(n+1)$ with $n \geq 2$ (equivalently, $\text{SU}(N_c)$ with $N_c \geq 3$).*

Proof. We test each family and each exceptional algebra against (F1)–(F3). The full data is:

Algebra	Group	$\dim_{\text{fund}}$	Type	(F1)	(F2)	Verdict
<i>Family $A_n = \mathfrak{su}(n+1)$</i>						
A_1	$\text{SU}(2)$	2	Pseudoreal	Fail	—	Excluded (F1)
A_2	$\text{SU}(3)$	3	Complex	Pass	$k=3$: Pass	Pass
A_3	$\text{SU}(4)$	4	Complex	Pass	$k=4$: Pass	Pass*
A_4	$\text{SU}(5)$	5	Complex	Pass	$k=5$: Pass	Pass*
$A_{n \geq 5}$	$\text{SU}(n+1)$	$n+1$	Complex	Pass	$k=n+1$: Pass	Pass*
<i>Family $B_n = \mathfrak{so}(2n+1)$, $n \geq 2$</i>						
B_2	$\text{SO}(5)$	5	Real	Fail	—	Excluded (F1)
B_3	$\text{SO}(7)$	7	Real	Fail	—	Excluded (F1)
$B_{n \geq 2}$	$\text{SO}(2n+1)$	$2n+1$	Real	Fail	—	Excluded (F1)
<i>Family $C_n = \mathfrak{sp}(2n)$, $n \geq 3$</i>						
C_3	$\text{Sp}(6)$	6	Pseudoreal	Fail	—	Excluded (F1)
$C_{n \geq 3}$	$\text{Sp}(2n)$	$2n$	Pseudoreal	Fail	—	Excluded (F1)
<i>Family $D_n = \mathfrak{so}(2n)$, $n \geq 4$ (vector and spinor carriers)</i>						
D_4	$\text{SO}(8)$	8	Real	Fail	—	Excluded (F1)
$D_{n \geq 4}$	$\text{SO}(2n)$	$2n$	Real	Fail	—	Excluded (F1)
D_4 spinor	$\text{Spin}(8)$	8	Real	Fail	—	Excluded (F1)
D_{2k} spinor	$\text{Spin}(4k)$	2^{2k-1}	Real/psr.	Fail	—	Excluded (F1)
D_5 spinor	$\text{Spin}(10)$	16	Complex	Pass	no cubic inv.	Excluded (F2) [†]
D_{2k+1} spinor	$\text{Spin}(4k+2)$	2^{2k}	Complex	Pass	[‡]	Excluded (F2) [‡]
<i>Exceptional algebras</i>						
G_2		7	Real	Fail	—	Excluded (F1)
F_4		26	Real	Fail	—	Excluded (F1)
E_6		27	Complex	Pass	$k=3$: sym., Fail	Excluded [†]
E_7		56	Pseudoreal	Fail	—	Excluded (F1)
E_8		248	Real	Fail	—	Excluded (F1)

Pass: satisfies (F1)–(F3) but is non-minimal (higher $\dim(G)$ than $\text{SU}(3)$; eliminated in Step 3 below).

[†] E_6 satisfies (F1), but its celebrated rank-3 invariant on the **27** is the *symmetric* cubic form, not a totally antisymmetric 3-tensor — so it does not satisfy (F2) as now correctly stated (the v3.9 table’s “ $k=3$: Pass” conflated the two symmetry types). Independently of (F2), E_6 is excluded by capacity minimality: $\dim(E_6) = 78$ vs. $\dim(\text{SU}(3)) = 8$; the cost cut is retained as the exclusion of record so that nothing rests on the invariant-type correction alone.

[‡]The $\text{Spin}(10)$ spinor **16** is a faithful complex carrier and passes (F1); it fails (F2) at low rank because it admits *no* cubic invariant at all: $\mathbf{16} \otimes \mathbf{16} = \mathbf{10} \oplus \mathbf{120} \oplus \mathbf{126}$ contains no $\overline{\mathbf{16}}$, so no invariant exists in $\mathbf{16}^{\otimes 3}$ of any symmetry type. Higher-rank antisymmetric singlets, where they exist, are large-body volume-form-type composites with no small- k screening role, and the cost ranking ($\dim \text{Spin}(10) = 45$ vs. 8) eliminates the candidate independently. The same two-step exclusion (no low-rank antisymmetric invariant; cost) covers the general D_{2k+1} spinor.

Mechanism of exclusion. The B_n , C_n families, the D_n vector carriers, the D_{2k} spinors, and the exceptionals G_2 , F_4 , E_7 , E_8 are all excluded by (F1) alone: those carriers are real or pseudoreal, and $B1'$ (Theorem 3.6 of Technical Supplement II (*The Foundational Gauge Program*)) proves that real/pseudoreal carriers cannot sustain oriented composites. The complex carriers among compact simple Lie algebras are the A_n ($n \geq 2$) fundamentals (and their conjugates — the conjugate fundamental is a second faithful $n+1$ -dimensional irrep, equivalent for this scan under the CPT/conjugation quotient), the E_6 **27**, and the D_{2k+1} chiral spinors; the latter two fall to (F2) and cost as recorded in the table notes.

E_6 passes (F1), fails (F2) (its rank-3 invariant on the **27** is symmetric, per the table), and is independently excluded on capacity grounds: $\dim(E_6) = 78$ requires 78 gauge generators, each costing one admissibility channel. The $\text{SU}(3)$ alternative requires 8 generators. By L_{Cauchy} (Theorem 1.3 of Technical Supplement II (*The Foundational Gauge Program*)), the cost functional is linear in generator count, so E_6 costs $78/8 = 9.75$ times as much as $\text{SU}(3)$. Minimality selects the cheapest adequate structure — and minimality is $A1 + \text{MD} + A2$, not $A1$ alone: $A1$ bounds the budget, MD (via L_{ε^*}) prices the floor, $A2$ selects the argmin (Lemma 2.24 of Technical Supplement II (*The Foundational Gauge Program*)).

The only surviving family is $A_n = \text{SU}(N_c)$ with $N_c \geq 3$. □

Where the E_6 cut actually happens. E_6 satisfies (F1) — it has a complex faithful carrier — but not (F2): its rank-3 invariant on the **27** is symmetric, not antisymmetric. The point of this paragraph survives that verdict: E_6 's exclusion does not *need* (F2). Independently of any representation-theoretic filter, the difference between “ $\text{SU}(N_c)$ survives” and “ E_6 is rejected” is a capacity-minimality result under the minimality package ($A1 + \text{MD} + A2$). A reader who doubts the invariant-type verdict on the cubic — or who accepts the carrier requirements but doubts the capacity-minimality clause — still faces the cost cut: the minimality clause closes the gap on its own, consumed at its grade (CM, Lemma 2.24 of Technical Supplement II (*The Foundational Gauge Program*)) through the admissibility-cost ledger. The E_6 exclusion is therefore a direct consequence of the minimum-cost ontology: E_6 is admissible (its cost fits below the capacity bound) but is not minimum-cost; $\text{SU}(3)$ is. “ E_6 is rejected” is shorthand for “ E_6 has higher cost than $\text{SU}(3)$ and both are admissible,” not for an act of rejection.

4.3 Capacity optimization: $N_c = 3$

Within the surviving $\text{SU}(N_c)$ family, all $N_c \geq 3$ pass the purely local color filters. The remaining question is not whether larger color groups are mathematically possible; they are. The APF question is which member of the admissible family minimizes the finite admissibility ledger under the same representation and anomaly constraints.

Definition 4.3 (Color optimization problem). The color optimization problem is the quadruple $(\mathcal{N}, J, \mathcal{C}, \preceq)$ where:

- (i) $\mathcal{N} = \{N_c \in \mathbb{N} : N_c \geq 3\}$ is the family surviving oriented-composite robustness ($B1'$).
- (ii) $\mathcal{C}(N_c)$ is the set of chiral templates with one color-doublet family, one lepton-doublet family, anomaly cancellation, asymptotic freedom, and the F1–F7 scan constraints generalized from 3 to N_c colors.
- (iii) The objective is the channel ledger

$$J(N_c) = (4N_c + 3)N_{\text{gen}} + 4 + ((N_c^2 - 1) + 3 + 1), \quad N_{\text{gen}} = 3,$$

where the three terms are fermions, Higgs, and gauge generators.

- (iv) $N_c \preceq N'_c$ iff $J(N_c) \leq J(N'_c)$ after both candidates pass the same template filters.

The optimization is invariant under positive affine rescalings of the ledger because L_{Cauchy} fixes all channel costs to the same integer unit; no continuous weight can reverse the ordering. The scope of J is clause (ii)’s template class: $J(N_c)$ proves family-monotonicity within SM-family-shaped templates (one color-doublet family, one lepton-doublet family, filters generalized to N_c colors); it is not a joint optimization over template shape. The joint statement is the two-phase reading below (Proposition 4.5 and the consistency note following it).

Remark 4.4 (What the color cost does and does not count). The objective J is a structural channel ledger, not an on-shell degree-of-freedom count and not a quadratic-Casimir energy functional. It counts three positive-cost resources: matter channels, Higgs channels, and gauge-generator routing channels. Casimir weights, beta-function coefficients, and propagating polarizations may refine dynamics after the template is fixed; they are not subtracted from the structural existence cost. A cost functional that subtracts gauge constraints, treats new generators as free, or cancels matter channels against ghosts is therefore not a small deformation of J but a different ontology: it counts physical excitations after gauge fixing rather than admissibility infrastructure before gauge fixing. Proposition 4.7 states the allowed robustness cone for genuine positive-cost deformations.

Two further filters apply:

Proposition 4.5 (Witten anomaly excludes even N_c). *For gauge group $\text{SU}(N_c) \times \text{SU}(2) \times \text{U}(1)$, the number of $\text{SU}(2)$ doublets per generation is $N_c + 1$ (one quark doublet contributing N_c colors, plus one lepton doublet). The Witten anomaly [5] requires this number to be even. Therefore N_c must be odd.*

Joint consistency with T_{field} . The statement “one quark doublet plus one lepton doublet per generation” is the SM-like template structure that emerges from T_{field} (§5) rather than an input independent of it. The Witten-anomaly cut applied in the current section therefore functions as a *joint consistency* check with the fermion scan: for any odd $N_c \geq 3$, the minimum-DOF anomaly-free template has the (quark-doublet, lepton-doublet) structure, and the Witten filter is satisfied; for any even N_c , no admissible template of this structure survives. The logical ordering is not “ N_c is fixed in §5 then templates are scanned in §6.” Rather, the admissible N_c ’s and admissible fermion templates are a joint optimum, and the supplement exhibits that joint optimum by first narrowing N_c to odd values under the generic template ansatz (this section) and then closing $N_c = 3$ by the full scan (§5). Readers who prefer a single-step derivation may take Propositions 4.5 and 4.6 together with Theorem 5.8 as a simultaneous result: ($N_c = 3$, SM template) is the unique joint minimum-cost admissible pair.

Proposition 4.6 (Capacity minimality selects $N_c = 3$). *Under Definition 4.3, within the SM-family template class of its clause (ii), $N_c = 3$ is the unique minimum-cost admissible color rank.*

Proof. The Witten anomaly removes even N_c . For odd $N_c \geq 3$ the objective is

$$J(N_c) = (4N_c + 3) \cdot 3 + 4 + (N_c^2 + 3) = N_c^2 + 12N_c + 16.$$

For $N'_c > N_c \geq 3$,

$$J(N'_c) - J(N_c) = (N'_c - N_c)(N'_c + N_c + 12) > 0.$$

Therefore J is strictly increasing on the admissible odd ranks, and $N_c = 3$ is the unique minimum. The first alternatives make the ordering visible:

Candidate	Local color filter	Witten filter	$J(N_c)$	Verdict
SU(2)	Fails $B1'$ / baryon-orientation rank	–	–	Not in $\mathcal{N}$
SU(3)	Pass	Pass (4 doublets)	61	Unique minimum
SU(4)	Pass local color	Fails (5 doublets)	–	Global anomaly
SU(5)	Pass	Pass (6 doublets)	101	Admissible but non-minimal
SU(7)	Pass	Pass (8 doublets)	149	Admissible but non-minimal
E_6	Simple alternative	No product chirality ledger	> 61	Non-minimal / wrong template

The result is therefore not a hidden phenomenological choice of three colors. It is the first admissible rank after the oriented-composite lower bound and the global anomaly cut, and the ledger is strictly monotone thereafter. $\square$

Proposition 4.7 (Robustness cone for the $N_c = 3$ argmin). *The $N_c = 3$ result is stable under every positive cost deformation that keeps fermion channels and gauge generators as independently costly admissibility resources. More precisely, replace Definition 4.3's unit ledger by*

$$J_{\mathbf{w}}(N_c) = w_g(N_c^2 + 3) + w_f(3(4N_c + 3)) + 4w_H + \rho(N_c), \quad (4)$$

with $w_g, w_f, w_H > 0$. If the perturbation satisfies

$$|\rho(N'_c) - \rho(N_c)| < (N'_c - N_c)[w_g(N'_c + N_c) + 12w_f] \quad (5)$$

for all odd $N'_c > N_c \geq 3$, then $N_c = 3$ remains the unique minimum among admissible odd ranks.

Proof. For any odd $N'_c > N_c \geq 3$,

$$J_{\mathbf{w}}(N'_c) - J_{\mathbf{w}}(N_c) = (N'_c - N_c)[w_g(N'_c + N_c) + 12w_f] + \rho(N'_c) - \rho(N_c). \quad (6)$$

The stated bound makes this difference strictly positive. Even ranks are removed by Proposition 4.5; $N_c = 2$ is removed by the oriented-composite carrier filter; exceptional/simple alternatives add independent generators or fail the same low-energy product-template filters. Thus the first admissible odd rank is the unique optimum throughout the positive robustness cone. The unit-cost ledger used in Definition 4.3 is the special case $w_g = w_f = w_H = 1$ and $\rho = 0$. $\square$

Remark 4.8 (Counterfactuals that would flip the color optimum). The theorem also identifies what would have to fail. A larger color rank could win only if one allowed negative or zero marginal cost for new gauge generators, treated additional colored fermion channels as cost-free, or changed the admissibility filters so that the $N_c = 3$ SM template was no longer admissible while a higher-rank template remained admissible. Those are not small perturbations of the APF ledger; they are changes to the resource semantics or to the physical constraints R1–R3.

4.4 The uniqueness core: abelian factor, center quotient, color rank

The preceding propositions establish the four pieces that reviewers most often conflate: algebra-level inner automorphisms, the linear cover acting on matter, the single relative abelian direction, and the color-rank optimum. For reference:

- U1** Algebraically, $\text{Inn}(\mathcal{A}) = \prod_k \text{PU}(n_k)$.
- U2** On charged matter modules, the relevant projective factors are represented by their simply connected covers, giving the familiar $\text{SU}(n)$ action where fundamental matter exists.
- U3** The center-action table in Proposition 2.26 of Technical Supplement II (*The Foundational Gauge Program*) shows that the common kernel on the derived multiplets is exactly $\mathbb{Z}_6$.
- U4** The color-rank optimum is the joint optimum over rank and template, summarized in Table 1.

Rank	Best local template at that rank	Minimum ledger	Reason not selected / selected
$N_c = 2$	two-color pseudo-real template	below SM-like count but fails baryon-orientation and chiral color separation	Excluded before optimization: no oriented three-color baryon carrier and no faithful QCD-like confinement ledger.
$N_c = 3$	SM template: Q, u^c, d^c, L, e^c	$J = 61$	Unique joint survivor: Witten-even doublet count, anomaly closure, minimum active channels.
$N_c = 4$	nearest SM-like extension	–	Witten anomaly: $N_c + 1 = 5$ weak doublets.
$N_c = 5$	SM-like extension with five colors	$J = 101$	Anomaly-compatible but strictly higher active-channel cost than $N_c = 3$.
$N_c \geq 7$ odd	SM-like extension	$J = N_c^2 + 12N_c + 16 > 101$	Strictly monotone cost above the $N_c = 3$ optimum.
Exceptional/simple unification	E_6 , $\text{SU}(5)$ -like simple carriers	> 61 or wrong low-energy factorization	Adds unneeded generators/exotics or imports high-energy unification structure not forced by finite-interface admissibility.

Table 1: Per- N_c joint optimum audit. The table distinguishes the earlier marginal SM-like comparison from the stronger joint statement: for each allowed color-rank family, the best admissible template either fails a structural filter or has larger active-channel ledger than ($N_c = 3$, SM).

Lemma 4.9 (Outer-permutation locality criterion). *Let $\mathcal{A} = \bigoplus_k M_{n_k}(\mathbb{C})$ be the finite admissibility algebra. A block permutation is a continuous gauge symmetry only if it preserves every interface’s local-support assignment and lies in the identity component of the codespace-preserving automorphism group. Outer permutations between distinct blocks fail this criterion: they move matter modules between support blocks, or, when the blocks are cost-degenerate, form a discrete permutation component separated from $\text{Inn}(\mathcal{A})$. Consequently they are kinematic/flavor relabelings, not Lie-algebra gauge generators.*

Proof. Inner automorphisms act inside a fixed simple summand and preserve the local support projector of each interface. A nontrivial outer permutation changes the summand label. If the summands are not identical as cost-equipped local codespaces, support preservation fails immediately. If they are identical, the permutation is an element of a finite symmetric group of degenerate labels; it has no infinitesimal generator and no continuous path to the identity inside $\text{Aut}(\mathcal{A})$. Gauge bosons arise from the Lie algebra of continuous local symmetries, so such discrete permutations cannot be gauge directions. $\square$

Edge case: family-non-universal $\text{U}(1)$ directions. Anomaly-free combinations such as $L_\mu - L_\tau$

require generation labels to be independently gauge-addressable distinctions. The minimal template used in this supplement treats generations as repeated copies of the same one-generation module; it does not pay an additional admissibility channel for “muon family” versus “tau family” as a primitive gauge label. Therefore family-non-universal abelian directions are excluded within the minimal-template theorem by capacity minimality, not by anomaly algebra alone. A theory that postulates generation-resolving structure as input lies outside the minimal APF theorem rather than contradicting it.

4.5 Chiral factor classification (R2)

Proposition 4.10 (SU(2) is the unique chiral factor). *R2 requires a compact simple Lie group with a faithful pseudoreal representation of dimension exactly 2. Among all compact simple Lie algebras:*

- (i) $\mathfrak{su}(2)$ has a 2-dimensional irreducible representation (the fundamental doublet) which is pseudoreal ($J = i\sigma_2 K$, $J^2 = -1$).
- (ii) All other compact simple Lie algebras have minimum faithful dimension ≥ 3 : $\mathfrak{su}(n+1)$ for $n \geq 2$ has $\dim_{\text{fund}} = n+1 \geq 3$; $\mathfrak{so}(2n+1)$ has $\dim_{\text{fund}} \geq 5$; $\mathfrak{sp}(2n)$ for $n \geq 2$ has $\dim_{\text{fund}} \geq 4$; all exceptionals have $\dim_{\text{fund}} \geq 7$.

Therefore SU(2) is the unique compact simple Lie group satisfying R2.

At the group level, the gauge factor must be SU(2) rather than its quotient $\text{SO}(3) = \text{SU}(2)/\mathbb{Z}_2$: the doublet representation does not descend to SO(3) (it is not single-valued under 2π rotations), so faithfulness requires the simply connected cover SU(2).

4.6 Abelian factor and simple-group alternatives (R3)

Proposition 4.11 (U(1) is the abelian carrier). *R3 requires at least one abelian grading direction, and $\text{U}(1) \cong \mathbb{R}/\mathbb{Z}$ is the unique connected compact 1-dimensional abelian Lie group, so each abelian grading direction is a U(1) (Theorem R, R3 derivation in §3 of Technical Supplement II (The Foundational Gauge Program)). That exactly one such factor closes the template — that a single U(1) with distinct hypercharge assignments resolves all multiplet degeneracies of the scanned content, and that any additional factor adds a gauge generator with no admissibility gain — is the a-posteriori closure of §7, consumed after the scan (Theorem 4.13, clause (ii)).*

Proposition 4.12 (Simple-group alternatives are excluded). *Any simple compact Lie group G_{simple} containing $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ as a subgroup has $\dim(G_{\text{simple}}) \geq 24$:*

Group	dim	Cost ratio	Contains SM?
$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$	12	1.0×	(Product)
SU(5)	24	2.0×	Yes (Georgi–Glashow)
SO(10)	45	3.75×	Yes
E_6	78	6.5×	Yes

Every simple envelope costs at least $2\times$ the product structure. Minimality (A1 + MD + A2, Lemma 2.24 of Technical Supplement II (The Foundational Gauge Program)) selects the product. The earlier versions added a second exclusion clause: that a simple group has a single coupling constant and therefore cannot yield independently running confining and chiral factors. That clause is withdrawn — it is not a valid exclusion, because symmetry breaking and threshold matching give the unbroken subgroup factors distinct infrared runnings from a single ultraviolet coupling (the standard grand-unification pattern). What survives, and suffices, is the cost comparison above:

every simple envelope prices at least $2\times$ the product structure at the same enforcement content, and minimality ($A1 + MD + A2$) selects the product. The requirement that the chiral factor not confine at accessible scales is dynamical input, not a group-theoretic exclusion, and is not used here.

4.7 Consolidation: unique gauge template

Combining Steps 1–6:

Theorem 4.13 ($L_{\text{gauge,unique}}$: Gauge Template Uniqueness). *Two clauses, at different logical stations.*

(i) *Nonabelian carrier ranking (forward). The unique nonabelian gauge structure satisfying the carrier requirements R1–R2 at minimum capacity cost is*

$$G_{\text{na}} = \text{SU}(3) \times \text{SU}(2),$$

and R3 requires at least one abelian grading direction alongside it. No alternative compact nonabelian structure satisfies both carrier requirements. No simple-group envelope is capacity-competitive.

(ii) *Abelian closure (a-posteriori pointer). That exactly one U(1) closes the template — necessity, sufficiency, directional uniqueness, and the inadmissibility of any second factor — is not part of the forward statement: it is proved a-posteriori for the selected matter content by the closure theorems of §7, consumed after the scan. The composite of (i) with that closure is the full gauge template*

$$G = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1), \quad \dim(G) = 12.$$

Derivation chain: $A1 \rightarrow \text{Theorem R (R1+R2+R3)} \rightarrow \text{Lie classification} \rightarrow \text{SU}(N_c) \times \text{SU}(2) (+ \text{ one abelian grading})$
 $\text{Witten} + \text{minimality} \rightarrow N_c = 3 \rightarrow \text{one-U}(1) \text{ closure (\S7)}.$

Imported mathematics: The classification of compact simple Lie algebras (Killing 1888, Cartan 1894) and the Witten anomaly theorem (Witten 1982) are used as external mathematical inputs. Both are established results with independent proofs; neither is framework-specific. All physical inputs (R1–R3) are Theorem R’s carrier requirements, proved at their grades in §3 of Technical Supplement II (The Foundational Gauge Program).

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³Code reference: `check_L_gauge_template_uniqueness` in `gauge.py`.

5. Fermion content: complete scan of the declared space

Reproducibility contract for the 1,680-template scan

The scan is fully specified by four inputs: (i) the derived gauge template $SU(3) \times SU(2) \times U(1)$; (ii) the bounded representation list proved in §5.2; (iii) the generation count and multiplicity bounds stated in the search-space construction; and (iv) the seven exact filters F1–F7 below. The output is the set of templates surviving all seven filters, quotiented by CPT. No floating-point thresholds, fitted coefficients, or data-dependent choices enter. A reproduction must report the initial 1,680 candidates, the waterfall counts in §5.4 under the canonical F6 predicate (full-system non-degeneracy: 10 survivors pre-CPT, 5 post-CPT), the robustness counts under the two variant predicates (spectator reduction 8/4; uniform dimension 4/2; the predicate statement in §5.3 states all three and designates the canonical one), and the unique minimum-capacity 45-Weyl-fermion survivor. The standalone script and repository link in [20] are therefore not supplemental evidence for a different claim; they are the machine-readable form of the finite enumeration specified here.

Audit files required for independent reproduction. A complete release consists of: `scan_inputs.json` (representation bounds and equivalence relation), `scan_outputs.csv` (all 1,680 canonical templates), `filter_waterfall.csv` (F1–F7 survivor counts), `near_misses.csv` (templates failing only the last nontrivial filter), and `certificate.sha256` (hash of the ordered bundle). If any of these files is absent from a public release, the paper’s scan claim should be read as a mathematically specified enumeration rather than as independently reproduced software evidence. The theorem claim itself is the finite enumeration proved below; the repository is the machine-checkable certificate layer.

Proposition 5.1 (Boundedness of the declared enumeration). *The 1,680-template scan is complete relative to the declared representation list and multiplicity caps of §5.2: P1–P2 exclude the enumerated representation enlargements unconditionally, P3–P5 certify dominance against the named extension directions, and the 1,680 candidates exhaust the declared space. The caps themselves are declared scan inputs; that no admissible sub-45-DOF template exists outside them is an open proof obligation, not a claim of this proposition.*

Proof sketch. P1–P2 reduce the representation content unconditionally: a single field in an $SU(3)$ representation of dimension ≥ 10 , or a colored field in an $SU(2)$ representation of dimension ≥ 3 , exceeds the AF budget regardless of the rest of the template. P3 and P5 are dominance certificates against the named enlargement directions (colorless triplets; field types beyond the anomaly-closed core), and P4 is the two-colored-doublet class-dominance result (class minimum 54, Proposition 5.5). Within the declared caps, the 1,680 candidates are the exact finite quotient of the declared space (the closed-form count in §5.2). What this establishes is boundedness and dominance, not universal completeness: the caps are declared inputs, and their completeness is the named open obligation of the reproducibility contract. The public scan artifact is the machine-readable certificate of this bounded enumeration. $\square$

Descriptive reading of the scan. The enumeration in this section is *not* a physical process by which nature “chooses” the Standard Model fermion content from a space of candidates. The admissibility conditions (five exclusion proofs bounding the search space, plus seven sequential filters encoding anomaly cancellation, asymptotic freedom, and doublet–singlet content) define a predicate $\text{Adm}(\mathcal{T})$ on chiral fermion templates. The scan is an *algorithmic verification* that the set $\{\mathcal{T} : \text{Adm}(\mathcal{T})\}$ has a unique member up to the stated equivalences — namely the 45-Weyl-fermion SM template. The 1,680 distinct templates (4,680 ordered iterations) are artefacts of the verification procedure, not candidates in a dynamical competition. The claim of this section is therefore descriptive: “the admissibility predicate is satisfied by exactly one template,” not “the framework selects one template

from among 1,680.”

This section specifies the 1,680-template search space, defines the seven sequential filters, provides the survivor count after each filter, and states the boundedness result at its licensed form: P1–P2 exclude the enumerated representation enlargements unconditionally, P3–P5 certify dominance against the named extension directions, and the caps themselves are declared scan inputs whose completeness is an open proof obligation.

5.1 Template definition

A *chiral fermion template* is a finite multiset $\mathcal{T} = \{(r_i^{(3)}, r_i^{(2)})\}_{i=1}^{|\mathcal{T}|}$ where $r_i^{(3)}$ is an irreducible representation of SU(3) and $r_i^{(2)}$ is an irreducible representation of SU(2). Each entry represents a left-handed Weyl fermion field. The total number of Weyl degrees of freedom per generation is $\text{DOF} = \sum_i \dim(r_i^{(3)}) \cdot \dim(r_i^{(2)})$. The template describes one generation; the full content is $N_{\text{gen}} \times \text{DOF}$ with $N_{\text{gen}} = 3$ (derived in Paper 2, §9).

The search space is bounded by five closed-form exclusion proofs that restrict both the allowed representations and the number of field types.

5.2 Search space bounds: five exclusion proofs

Proposition 5.2 (P1: Large SU(3) representations are AF-excluded). *Any SU(3) representation with dimension ≥ 10 (i.e., **10**, **15**, ...) exceeds the SU(3) asymptotic freedom budget as a single field. For the **10**: the Dynkin index is $T(\mathbf{10}) = 15/2$; under the all-left-handed Weyl convention of the scan (coefficient $2/3$ per Weyl field — the $4/3$ form is the Dirac-pair coefficient of §2 and does not apply to Weyl entries), the AF contribution of a single field in the **10** with SU(2) singlet and 3 generations is $(2/3)(15/2)(1)(3) = 15 > 11$. Therefore no template containing any SU(3) rep of dimension ≥ 10 can be asymptotically free. The allowed SU(3) representations are: **1**, **3**, **$\bar{3}$** , **6**, **$\bar{6}$** , **8**.*

Proposition 5.3 (P2: Colored SU(2) triplets and higher are AF-excluded). *Any colored field ($\dim(r^{(3)}) \geq 3$) in an SU(2) representation of dimension ≥ 3 exceeds the SU(2) AF budget. For a **(3, 3)** field: $T(\mathbf{3}_{\text{SU}(2)}) = 2$, $\dim(\mathbf{3}_{\text{SU}(3)}) = 3$, Weyl-convention contribution = $(2/3)(2)(3)(3) = 12 > 22/3 \approx 7.33$. Therefore colored fields are restricted to SU(2) representations $\{\mathbf{1}, \mathbf{2}\}$.*

Remark (what P2 does and does not exclude). P2’s inequality carries the factor $N_{\text{gen}} = 3$: it excludes colored weak-triplets at the forced multiplicity. A single unreplicated copy of such a field is AF-clean ($(2/3)(2)(3)(1) = 4 < 22/3$), and contents built on it can be chiral and anomaly-free. Their exclusion is the multiplicity theorems of §8 (the capacity–beta identities and the rival kill), not this proposition alone. Example 4 of the worked-example section exhibits the sharpest such rival and its kill map.

Proposition 5.4 (P3: Colorless SU(2) triplets cannot lower the minimum). *This is a dominance bound, not an a-priori search-space theorem, and its logical form matters. Phase one: the scan, run on the declared template space (which contains no colorless triplets), exhibits an admissible survivor at 45 DOF. Phase two: any template extending or modifying a survivor by a colorless SU(2) triplet carries that triplet’s own $3 \times 3 = 9$ DOF (or, replacing a doublet, a net +3), so no colorless-triplet template beats the exhibited 45 without removing content that the filters (doublet–singlet content, anomaly closure) require. The proposition therefore certifies that enlarging the declared space by colorless triplets cannot lower the minimum; it is consumed after the scan’s phase-one result, never before it. (The v3.9 wording cited the SM minimum inside a search-space bound — circular as stated; the two-phase form above is the intended and valid argument.)*

Proposition 5.5 (P4: Multiple colored doublets are DOF-dominated). *Every template with two colored SU(2) doublets that passes the printed filters carries at least 54 Weyl degrees of freedom over three generations — strictly above the 45-DOF Standard-Model template. The class minimum is attained by the conjugate-pair template*

$$(\mathbf{3}, \mathbf{2})_{Y_1} + (\bar{\mathbf{3}}, \mathbf{2})_{-Y_1} + (\mathbf{3}, \mathbf{1})_{Y_2} + (\bar{\mathbf{3}}, \mathbf{1})_{-Y_2}, \quad \text{DOF} = (6 + 6 + 3 + 3) \times 3 = 54,$$

which passes every printed filter: $b_0^{(3)} = 5 > 0$ (F1), $b_0^{(2)} = 4/3 > 0$ (F2), the F3 content predicate (both types present), exact $[\text{SU}(3)]^3$ cancellation (F4), six doublets counted with color multiplicity — even (F5), and the full anomaly system, which vanishes identically for any rational (Y_1, Y_2) with the conjugate structure, non-degenerately for $Y_1 \neq 0$ (F6). The class bound is a direct enumeration of the two-colored-doublet class under the printed filters, in exact rational arithmetic, carried in the release’s audit layer. Two earlier claims are withdrawn as false: the premise that two colored doublets force lepton content (the conjugate-pair template is anomaly-free with no leptons at all), and the 63-DOF “tightest template” (its seven doublet components, counted with color multiplicity, are odd — it fails the Witten filter F5 and was never admissible). What P4 contributes to the search-space bound is dominance: $54 > 45$, so no two-doublet template can beat the Standard-Model minimum.

Proposition 5.6 (P5: Extra field types cannot lower the minimum). *Same dominance form as P3. The a-priori per-type floor (≥ 3 DOF: a colorless singlet across 3 generations) does not by itself bound many-type templates near 45 — six minimal types floor at only 18 DOF, and the v3.9 wording (“ $\geq 45 + 3$ ”) presupposed the five-type baseline it was meant to establish. The valid statement: within the declared space, every F6 survivor containing types beyond the five SM roles carries them as additions to an anomaly-closed core (the scan’s survivor list exhibits exactly this — the 6- and 7-entry survivors sit at 48–72 DOF, each strictly above the 45-DOF core it extends), so extra types never lower the minimum. Completeness of the declared space itself is a separate obligation, stated with the enumeration below.*

P1–P2 are theorems bounding the representation content (AF-forced): SU(3) reps $\in \{3, \bar{3}, 6, \bar{6}, 8\}$ (excluding singlets, which are colorless) and, for colored fields, SU(2) reps $\in \{1, 2\}$. The remaining bounds are *declared enumeration caps*, stated here as scan inputs rather than theorems: one colored SU(2) doublet (P4 certifies dominance against the two-doublet enlargement direction), at most 3 colored singlets, at most one lepton doublet, at most 2 lepton singlets — hence template size $|\mathcal{T}| \leq 7$ (the v3.9 text said $|\mathcal{T}| \leq 5$, which contradicted its own enumeration: the survivor list contains 6- and 7-entry templates). Completeness of these caps — that no admissible sub-45-DOF template exists outside them — is carried by P3/P5-style dominance for the enumerated directions and is otherwise an explicit proof obligation of the reproducibility contract, not a closed theorem. Combined with multiplicity choices (0–3 colored singlets drawn with replacement, 0–2 lepton singlets, doublet present or absent), this yields 1,680 distinct candidate templates.⁴

5.3 Seven filters

Each filter is an exact declared predicate, stated at its ledger type in Theorem 5.9: F1–F2 consume the imported one-loop Weyl coefficient and the imported generation count (C3–C4); F3 is the doublet–singlet content predicate (C6); F4–F6 are the imported anomaly and Witten equations consumed as

⁴Colored singlets are unordered within a template, so the correct enumeration uses combinations with replacement: $5 \times (1 + 5 + 15 + 35) \times 2 \times 3 = 1,680$. A standalone verification script with exact rational arithmetic and four built-in red-team challenges is archived at <https://doi.org/10.5281/zenodo.19154197>; an interactive Colab walkthrough is at https://colab.research.google.com/drive/1ot01Qwp_Lr30sK6ykkeMRATanecPkKg6; source code is at <https://github.com/Ethan-Brooke/APF-Paper-2-The-Structure-of-Admissible-Physics>.

filter conditions (C5), with F6 additionally carrying the independent electromagnetic-completeness assumption in its canonical full-system form (C8); F7 is the CPT convention (C7). Their APF motivations live where cited — the asymptotic-freedom discussion of §2 and the foundational program of Technical Supplement II (*The Foundational Gauge Program*) — and the classification consumes them as stated inputs. The filters are applied sequentially; each eliminates candidates that passed all previous filters.

- F1 SU(3) asymptotic freedom.** $b_0^{(3)} = 11 - (2/3) \sum_i T(r_i^{(3)}) \dim(r_i^{(2)}) \cdot N_{\text{gen}} > 0$, with the sum over the template’s left-handed Weyl entries (2/3 is the Weyl coefficient; the verifier has always used it — the 4/3 printed here in v3.9 was the Dirac-pair form, a display error that, with Weyl entries, mis-prices the SM at $b_0 = 3$ instead of 7). $N_{\text{gen}} = 3$ is imported (§8); F1–F2 verdicts are conditional on it. Derived from non-abelian self-interaction dominance (§2).
- F2 SU(2) asymptotic freedom.** $b_0^{(2)} = 22/3 - (2/3) \sum_i T(r_i^{(2)}) \dim(r_i^{(3)}) \cdot N_{\text{gen}} > 0$ (same Weyl convention and N_{gen} conditionality as F1).
- F3 Doublet–singlet content.** The template must contain both colored doublets and colored singlets. This is a content predicate, not a chirality test, and we name it at its content: co-presence of the two types removes doublet-only and singlet-only templates, but it does not by itself exclude vector-like content — the conjugate-pair two-doublet template of Proposition 5.5 contains both types and is exactly vector-like (every field acquires a gauge-invariant conjugate mass partner). Within the scanned space, which carries exactly one colored doublet by construction, no conjugate colored-doublet partner can occur, and the chirality of the surviving assignment is settled downstream, at the charge level, by F6’s non-degeneracy ($Y_Q \neq 0$). A true chirality predicate — the absence of any gauge-invariant rep-conjugate pairing with opposite hypercharge — is charge-dependent and evaluable only after hypercharges are assigned; it is reported separately in the release’s audit layer rather than consumed as a scan filter.
- F4 [SU(3)]³ anomaly cancellation.** $\sum_i A(r_i^{(3)}) \cdot \dim(r_i^{(2)}) = 0$, where A is the cubic anomaly coefficient ($A(\mathbf{3}) = 1/2$, $A(\bar{\mathbf{3}}) = -1/2$, $A(\mathbf{6}) = 7/2$, $A(\bar{\mathbf{6}}) = -7/2$, $A(\mathbf{8}) = 0$).
- F5 Witten anomaly freedom.** The total number of SU(2) doublets (counted with SU(3) multiplicity) must be even: $\sum_i \dim(r_i^{(3)}) \cdot [r_i^{(2)} = \mathbf{2}] \equiv 0 \pmod{2}$.
- F6 Full anomaly system solvable.** The combined system of $[\text{SU}(2)]^2[\text{U}(1)]$, $[\text{SU}(3)]^2[\text{U}(1)]$, $[\text{grav}]^2[\text{U}(1)]$, and $[\text{U}(1)]^3$ anomaly cancellation conditions must admit rational hypercharge solutions. Templates containing colored singlets with $A(R) = 0$ (e.g. the $\mathbf{8}$) are handled by spectator reduction: the spectator-reduced branch sets $Y = 0$ on such fields, decoupling them from the anomaly conditions, and the reduced template is then tested analytically. (The *full* anomaly system, solved without spectator reduction, additionally admits charged-octet solutions with $Y_8 = -\frac{3}{5} Y_Q$; they are eliminated at the minimality cut — see the predicate statement below.) For the spectator-free core, the system reduces to a quadratic in $z = Y_u/Y_Q$ having rational roots (§6). This reduction presupposes a hypercharged quark doublet ($Y_Q \neq 0$), and that non-degeneracy condition is load-bearing rather than incidental. Read weakly — as the bare existence of *some* anomaly-free assignment — F6 would admit a lighter spurious survivor: the Standard-Model generation stripped of its charged lepton singlet, $\{Q(\mathbf{3}, \mathbf{2}), \bar{\psi}_a(\bar{\mathbf{3}}, \mathbf{1}), \bar{\psi}_b(\bar{\mathbf{3}}, \mathbf{1}), L(\mathbf{1}, \mathbf{2})\}$ at $42 < 45$ Weyl degrees of freedom, which is chiral and passes F1–F5, and is anomaly-free under the degenerate assignment $Y_Q = Y_L = 0$, $Y = \pm t$ on the two colored singlets. The framework excludes it by **electromagnetic completeness** — every coloured multiplet must carry a separating charge, so $Y_Q \neq 0$, and the spectator-free anomaly system is therefore required to be *non-degenerate*, not merely

solvable. Electromagnetic completeness is hereby declared an *independent assumption* of the scan at exactly this point of use: it is motivated by the admissibility semantics (EC’s separating-label requirement applied to the unbroken electromagnetic charge), but it is not a consequence of anomaly cancellation, and it does the work of removing the 42-DOF near-miss. A reader who rejects it gets the stripped generation as a second survivor below the SM. The stripped template is recorded as the sharpest near-miss in `near_misses.csv`. *Predicate statement for re-implementers:* the **canonical** F6 predicate — the convention generating this document’s waterfall table and every count in the body — is *full-system non-degeneracy*: solve the complete anomaly system, spectators included, and require a non-degenerate rational solution ($Y_Q \neq 0$) — 10 survivors pre-CPT, 5 post-CPT. The full system admits charged-octet solutions, with $Y_8 = -\frac{3}{5} Y_Q$ forced linearly on every solution of the DOF-66 template (e.g. $(Y_Q, Y_u, Y_d, Y_8, Y_L) = (5, 13, -5, -3, -15)$), and SM-embedding solutions on the 69- and 72-DOF templates; all are eliminated at the minimality cut. Two variant predicates are in circulation across the artifacts and are reported as robustness checks with their counts (§5.4): (i) *spectator reduction* (implemented by the shipped standalone scan, v2, Zenodo 10.5281/zenodo.19154197): strip anomaly-free spectators ($A(R) = 0$ colored singlets, set to $Y = 0$ by the reduced system) and require a non-degenerate rational solution on the remainder — 8 pre-CPT, 4 post-CPT; (ii) *uniform dimension* (the in-tree code path: `check_T_field` in `gauge.py` requires every colored singlet to have SU(3) dimension $N_c = 3$) — 4 pre-CPT, 2 post-CPT. The winner (minimum DOF 45) is identical under all three predicates; only the intermediate counts differ⁵. The predicate inequivalence is witnessed by the DOF-66 charged octet: it survives the canonical full-system predicate and is absent under spectator reduction. One further computed fact sharpens the independence declaration above: EC-label-injectivity does *not* remove the 42-DOF near-miss — its unique nontrivial solution $(0, t, -t, 0)$ is label-injective and per-field faithful — so F6’s independence from EC is a computed theorem over this scan space, not only a declaration.

F7 CPT quotient. Templates related by charge conjugation (replacing every representation with its conjugate) describe the same physics. The CPT equivalence class is identified and only the canonical representative is counted.

5.4 Waterfall table

Filter	Survivors	Eliminated	Mechanism
Search space (P1–P5)	1,680	—	Bounded by exclusion proofs
F1+F2: Asymptotic freedom	348	1,332	$b_0^{(3)} > 0$ and $b_0^{(2)} > 0$
F3: Doublet–singlet content	324	24	Colored doublets + singlets required
F4: $[\text{SU}(3)]^3$ anomaly	24	300	$\sum A_3 d_2 = 0$
F5: Witten anomaly	12	12	#doublets even
F6: Full anomaly system	10	2	Full-system non-degenerate rational hypercharges ($Y_Q \neq 0$)
F7: CPT quotient	5	5	Conjugate pairs identified
Minimality	1	4	Min DOF = 45 selected

Counting convention (narrative vs. enumeration). The counts in the table above are the number of *distinct* templates surviving each filter, with unordered colored-singlet positions, under the canonical full-system F6 predicate. Both implementations (`check_T_field` in `gauge.py` and `fermion_scan_standalone.py`) use an ordered-tuple enumeration (`itertools.product`) and therefore test each distinct template multiple times, producing a raw iteration count of 4,680 rather

⁵Code reference: `check_L_F6_not_from_EC` in `ec_inventory_reading.py`.

than 1,680. Their F6 conventions are the two robustness variants: the standalone scan’s spectator reduction gives $8 \rightarrow 4$, and `check_T_field`’s uniform-dimension convention gives $4 \rightarrow 2$ after quotienting the ordering redundancy (the predicate statement in §5.3 states all three conventions and designates the canonical one). The minimality winner is identical under either enumeration and every stated F6 convention: the Standard Model at 45 DOF. A reviewer running the code will see the enumeration multiplicities and the robustness-variant counts; the table reports the unordered canonical count because that is what the five exclusion proofs (P1–P5) actually bound.

Robustness: the two variant F6 predicates. Templates #5–#10 in the post-F6 listing below include the adjoint **8** as a colored SU(2) singlet. Under the canonical full-system predicate these templates survive F6 — the DOF-66 pair via its charged-octet solutions ($Y_8 = -\frac{3}{5} Y_Q$), the 69- and 72-DOF pairs via their SM-embedding solutions — and fail only at the minimality cut. Two variant predicates are reported as robustness checks. *Spectator reduction*, implemented by the shipped standalone scan (`fermion_scan_standalone.py`), strips the anomaly-free spectator ($A(\mathbf{8}) = 0$, set to $Y = 0$ by the reduced system) and tests the remainder: 8 survivors pre-CPT, 4 post-CPT — the DOF-66 pair drops, because its spectator-reduced remainder is the degenerate 42-DOF stripped generation; the charged octet is exactly the predicate-inequivalence witness. *Uniform dimension*, the in-tree code path (`check_T_field` in `gauge.py`), requires every colored singlet at SU(3) dimension $N_c = 3$, eliminating octet templates at F6: 4 pre-CPT, 2 post-CPT. All three predicates reproduce the same winner — the SM at DOF 45 — at the minimality cut; the intermediate counts are 10/5 (canonical), 8/4, and 4/2.

The final winner is exact and reproduced by `check_T_field` in `gauge.py` and independently by `fermion_scan_standalone.py` [20].

Proposition 5.7 (Filter-order independence). *The survivor set $\mathcal{S} = \{\mathcal{T} : \mathcal{T} \text{ passes F1–F7}\}$ is independent of the order in which the seven filters are applied.*

Proof. Each filter F_k defines a predicate $P_k : \mathcal{T} \rightarrow \{0, 1\}$ on templates. A template survives all filters iff it satisfies the conjunction $P_1 \wedge P_2 \wedge \cdots \wedge P_7$. Logical conjunction is commutative and associative: $P_{\sigma(1)} \wedge \cdots \wedge P_{\sigma(7)} = P_1 \wedge \cdots \wedge P_7$ for any permutation $\sigma \in S_7$. Therefore the survivor set $\mathcal{S} = \{\mathcal{T} : \bigwedge_k P_k(\mathcal{T}) = 1\}$ is identical regardless of evaluation order.

What changes with filter order is the *waterfall count* (number of templates eliminated at each stage), because a template eliminated by F_k in the standard order might instead be eliminated by F_j with $j > k$ if the order is reversed. The final survivor set, however, is order-invariant. The standalone verification script confirms this under its spectator-reduction F6 predicate: running all $7! = 5,040$ permutations on the 1,680-template set yields the same 4 post-CPT survivors (and the same unique DOF-45 winner) in every case. The conjunction argument is predicate-independent, so the invariance holds equally for the canonical full-system predicate’s 5 post-CPT survivors. $\square$

Search-space bias and completeness. A distinct concern is whether the search space itself (the 1,680 templates bounded by P1–P5) introduces bias by excluding candidates that might survive the filters. P1–P2 are unconditional theorems: a single field in an SU(3) representation of dimension ≥ 10 , or a colored field in an SU(2) representation of dimension ≥ 3 , exceeds the AF budget regardless of what else the template contains, and the AF and DOF bounds are monotone (adding fields only increases DOF and shrinks the AF margin). The remaining bounds are declared enumeration caps (§5.2): P3 and P5 are dominance results certifying that enlarging the declared space in the enumerated directions cannot lower the minimum, and P4 is the two-colored-doublet class-dominance enumeration (class minimum 54, computed over the two-doublet class — outside the caps, which

admit one colored doublet). The licensed claim is therefore minimality *among the enumerated candidates*, together with dominance against the named enlargement directions. Completeness of the caps — that no admissible sub-45-DOF template exists outside them — is an explicit proof obligation of the reproducibility contract, not a closed theorem; the proof obligations named there are the open program.

Explicit listing of all post-F6 survivors. The 10 templates surviving F1–F6 under the canonical full-system predicate (before CPT quotient) are:

#	Template	DOF	Status
1	$(\mathbf{3}, 2) + (\bar{\mathbf{3}}, 1) + (\bar{\mathbf{3}}, 1) + (\mathbf{1}, 2) + (\mathbf{1}, 1)$	45	SM
2	$(\bar{\mathbf{3}}, 2) + (\mathbf{3}, 1) + (\mathbf{3}, 1) + (\mathbf{1}, 2) + (\mathbf{1}, 1)$	45	CPT of #1
3	$(\mathbf{3}, 2) + (\bar{\mathbf{3}}, 1) + (\bar{\mathbf{3}}, 1) + (\mathbf{1}, 2) + (\mathbf{1}, 1) + (\mathbf{1}, 1)$	48	SM + extra singlet
4	$(\bar{\mathbf{3}}, 2) + (\mathbf{3}, 1) + (\mathbf{3}, 1) + (\mathbf{1}, 2) + (\mathbf{1}, 1) + (\mathbf{1}, 1)$	48	CPT of #3
5	$(\mathbf{3}, 2) + (\bar{\mathbf{3}}, 1) + (\bar{\mathbf{3}}, 1) + (\mathbf{8}, 1) + (\mathbf{1}, 2)$	66	Charged octet ($Y_8 = -\frac{3}{5}Y_Q$)
6	$(\bar{\mathbf{3}}, 2) + (\mathbf{3}, 1) + (\mathbf{3}, 1) + (\mathbf{8}, 1) + (\mathbf{1}, 2)$	66	CPT of #5
7	$(\mathbf{3}, 2) + (\bar{\mathbf{3}}, 1) + (\bar{\mathbf{3}}, 1) + (\mathbf{8}, 1) + (\mathbf{1}, 2) + (\mathbf{1}, 1)$	69	SM + octet (SM-embedding)
8	$(\bar{\mathbf{3}}, 2) + (\mathbf{3}, 1) + (\mathbf{3}, 1) + (\mathbf{8}, 1) + (\mathbf{1}, 2) + (\mathbf{1}, 1)$	69	CPT of #7
9	$(\mathbf{3}, 2) + (\bar{\mathbf{3}}, 1) + (\bar{\mathbf{3}}, 1) + (\mathbf{8}, 1) + (\mathbf{1}, 2) + (\mathbf{1}, 1) + (\mathbf{1}, 1)$	72	SM + octet + singlet
10	$(\bar{\mathbf{3}}, 2) + (\mathbf{3}, 1) + (\mathbf{3}, 1) + (\mathbf{8}, 1) + (\mathbf{1}, 2) + (\mathbf{1}, 1) + (\mathbf{1}, 1)$	72	CPT of #9

Templates #5–#10 contain the adjoint **8** as a colored SU(2) singlet. Under the canonical full-system predicate they carry the anomaly solutions recorded in the predicate statement (the DOF-66 pair with the octet charged at $Y_8 = -\frac{3}{5}Y_Q$; the 69- and 72-DOF pairs with SM-embedding solutions). Their DOF cost ($8 \times 3 = 24$ per generation for the octet alone) is far above the minimum.

After the CPT quotient (F7), templates #2, #4, #6, #8, and #10 are identified with #1, #3, #5, #7, and #9 respectively, leaving 5 CPT-inequivalent survivors. Minimality selects DOF = 45, giving exactly one winner: the Standard Model — under F6 (electromagnetic completeness, an independent assumption, §5.3); the F6-free argmin is the 42-DOF stripped generation recorded there as the sharpest near-miss. The second-lightest survivor (#3, DOF = 48) differs from the SM by one additional right-handed neutrino-like singlet per generation; its exclusion is by minimality ($48 > 45$); a bridge-conditional observational cross-check is recorded at its corpus home (§9).

5.5 The unique survivor

Theorem 5.8 (T_{field} : SM Fermion Content). *The unique minimal chiral fermion template consistent with $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, three generations, electromagnetic completeness (F6 — an independent assumption, declared at its point of use in §5.3, in its canonical full-system non-degeneracy form), and all remaining admissibility constraints is:*

$$\{Q(\mathbf{3}, 2), u^c(\bar{\mathbf{3}}, 1), d^c(\bar{\mathbf{3}}, 1), L(\mathbf{1}, 2), e^c(\mathbf{1}, 1)\} \times 3 \text{ generations,}$$

with $\text{DOF} = (6 + 3 + 3 + 2 + 1) \times 3 = 45$ Weyl fermions.

No other template among the 1,680 candidates has $\text{DOF} \leq 45$ and passes all seven filters. P1–P2 exclude the enumerated representation enlargements unconditionally; P3–P5 certify dominance for the enumerated extension directions; the caps themselves are declared scan inputs whose completeness is an open proof obligation (§5.2). The licensed statement: the SM fermion content

is the unique minimum among the enumerated candidates, with dominance against the named enlargements.⁶; standalone verification: `fermion_scan_standalone.py` [20]

The scan result and its premise ledger assemble into the supplement’s citable core, stated as one theorem with the full input list typed — theorem, imported result, declared input, reading, or convention.

Theorem 5.9 (Conditional classification of the fermion template). *Fix the following premises — the full input ledger of this theorem.*

- (C1) **Declared scan inputs.** *The template caps and representation tables of §5.2 (P1–P2 as theorems; the enumeration caps as declared inputs).*
- (C2) **The gauge template.** *The fixed template $SU(3) \times SU(2) \times U(1)$, consumed through T_{field} ’s GIVEN (Theorem 5.8).*
- (C3) **Generation count.** *$N_{\text{gen}} = 3$, imported (§8) and consumed by the F1–F2 filters.*
- (C4) **One-loop Weyl beta coefficient.** *The coefficient $2/3$ per left-handed Weyl entry in the F1–F2 asymptotic-freedom filters, imported from standard gauge theory (§2).*
- (C5) **The anomaly system.** *The four gauge and gravitational anomaly equations and the Witten parity condition (F4–F6 and F5), consumed as filter conditions; their solutions are computed exactly over the rationals, not assumed.*
- (C6) **The F3 content predicate.** *F3’s doublet–singlet co-presence condition: colored doublets and colored singlets both present. A content predicate, not a chirality test (§5.3 states its honest role; chirality of the winner is settled at F6’s non-degeneracy).*
- (C7) **CPT identification.** *F7’s convention that conjugate templates describe the same physics.*
- (C8) **Electromagnetic completeness (F6).** *An independent assumption, declared at its point of use (§5.3), in its canonical full-system non-degeneracy form ($Y_Q \neq 0$ on the complete anomaly system); without it the argmin moves to the 42-DOF stripped generation. F6 and EC-label-injectivity are logically independent — computed, in both directions⁷.*
- (C9) **The objective.** *Weyl-component count at the left-handed Weyl convention: the quantity minimized is total Weyl degrees of freedom.*
- (C10) **Cost convention.** *Equal unit cost per counted channel — one capacity slot of equal weight per Weyl component, Higgs component, and gauge generator (BA5, §3; L_{Cauchy} ’s unit normalization $F(n) = n$) — with the gauge-cost equality $C(G) = \dim(G)$ resting on the named operational premise pair (operational encoding; disjoint-anchor realizability; Proposition 4.1).*

Under (C1)–(C10), the Standard-Model five-multiplet template at 45 Weyl degrees of freedom is the unique minimum among the enumerated candidates, computed exactly⁸.

Two remarks fix the premises’ status honestly. First, within APF the premises (C2) and (C3) have banked grounds — the gauge-uniqueness chain (§4, resting on the foundational program’s carrier theorems) and the capacity–beta identities with their coderefs (§8) — but this theorem consumes them as inputs: its truth does not certify them, and their grounds live where cited. Second, the abelian-factor count is likewise not certified here. The one- $U(1)$ results (§7) are a-posteriori closure theorems for the selected template: they consume the template and the scanned matter content and verify consistency and minimal-label uniqueness of the abelian sector, under the reading R-EC-inv with the entry-typed inventory I-typing⁹; neither this theorem nor those derive the other’s input. Of the premises: (C8) is an independent assumption; (C1)’s caps, (C7), (C9), and (C10) are declared

⁶Code reference: `check_T_field` in `gauge.py`.

⁷Code reference: `check_L_F6_not_from_EC` in `ec_inventory_reading.py`.

⁸Code reference: `check_T_field` in `gauge.py`.

⁹Code reference: `check_L_EC_inventory_reading` in `ec_inventory_reading.py`.

conventions of the enumeration and the ledger; (C4) and (C5) are imported standard results; (C2) and (C3) are imported APF results consumed as inputs.

The theorem statements upstream and downstream of the scan (Theorem 5.8, the waterfall conclusion of §5.4, Theorem 7.7) carry their conditionality clauses individually; this consolidated statement is the one to cite.

6. Anomaly cancellation and hypercharge uniqueness

Novelty boundary: standard anomaly algebra versus APF input selection

The anomaly equations used in this section are standard Standard-Model consistency equations. The APF novelty is not the algebraic fact that, given the SM chiral template, anomaly cancellation fixes hypercharge up to normalization. The novelty claim is upstream: the finite admissibility argument selects the gauge template and the anomaly-tested fermion template before the standard equations are applied. Thus the anomaly calculation is a closure check on APF-selected inputs, not an independent new anomaly formalism.

What is standard and what is new. The anomaly equations in this section are standard four-dimensional gauge anomaly conditions applied to a chiral fermion template. APF does not claim novelty for the algebraic fact that, once the SM representation pattern is assumed, anomaly cancellation fixes a rational hypercharge ray (up to normalization and charge conjugation). The APF contribution is upstream: the gauge template, the representation pattern, the absence of exotic fermions, the single abelian factor, the generation count entering the ledger, and the minimum-capacity choice of the 45-Weyl survivor are all fixed before the anomaly system is solved. Hypercharge uniqueness is therefore a sanity check and closure theorem for the derived template, not a disguised input of the template.

6.1 The four anomaly conditions

The quantum consistency of a chiral gauge theory requires the cancellation of all gauge anomalies. For $SU(3) \times SU(2) \times U(1)$ with the surviving template, there are four independent conditions on the five hypercharges $\{Y_Q, Y_u, Y_d, Y_L, Y_e\}$. Three are linear:

$$[SU(2)]^2[U(1)] = 0 : \quad N_c Y_Q + Y_L = 0, \quad (7)$$

$$[SU(3)]^2[U(1)] = 0 : \quad 2Y_Q - Y_u - Y_d = 0, \quad (8)$$

$$[\text{grav}]^2[U(1)] = 0 : \quad 2N_c Y_Q + 2Y_L - N_c Y_u - N_c Y_d - Y_e = 0. \quad (9)$$

The fourth is the cubic anomaly:

$$[U(1)]^3 = 0 : \quad 2N_c Y_Q^3 + 2Y_L^3 - N_c Y_u^3 - N_c Y_d^3 - Y_e^3 = 0. \quad (10)$$

6.2 Reduction to a quadratic

Solving (7)–(9) gives $Y_L = -N_c Y_Q$, $Y_d = 2Y_Q - Y_u$, $Y_e = -2N_c Y_Q$. Substituting into (10) and dividing by $Y_Q^3 \neq 0$ (non-trivial grading), define $z := Y_u/Y_Q$:

$$z^2 - 2z - (N_c^2 - 1) = 0. \quad (11)$$

Why the roots are rational. Equation (11) is a *monic* polynomial (leading coefficient = 1) with integer coefficients (-2 and $-(N_c^2 - 1)$, where $N_c \in \mathbb{Z}$). By the rational root theorem for monic integer polynomials: if $z = p/q$ in lowest terms is a root, then $q \mid 1$, so $z \in \mathbb{Z}$. The discriminant $\Delta = 4 + 4(N_c^2 - 1) = 4N_c^2$ is a perfect square, confirming integer roots:

$$z = 1 \pm N_c. \quad (12)$$

Since $Y_L = -N_c Y_Q$, $Y_d = 2Y_Q - Y_u = (2 - z)Y_Q$, and $Y_e = -2N_c Y_Q$ are all integer multiples of Y_Q , and Y_Q itself is a free normalization (any nonzero rational value is admissible, since it sets the

overall scale of the $U(1)$ generator; the compact- $U(1)$ representation-data premise of the front-matter ledger), choosing $Y_Q = 1/6$ gives $Y_i \in \mathbb{Q}$ for all five multiplets. What the anomaly system proves is that all hypercharge *ratios* are rational — the solution is a unique rational ray, up to the $u \leftrightarrow d$ relabeling. The absolute unit is fixed by the compactness of the $U(1)$ and the character-lattice convention ($Y_Q = 1/6$; the compact- $U(1)$ representation-data premise of the front-matter ledger), not by anomalies alone. For $N_c = 3$: $z \in \{4, -2\}$. These are related by $z \rightarrow 2 - z$ (the $u \leftrightarrow d$ relabeling), so they describe the same physics. Choosing $z = 4$ and normalizing $Y_Q = 1/6$:

Theorem 6.1 ($L_{\text{hypercharge}}$: Unique Hypercharge Assignment).

$$Y_Q = \frac{1}{6}, \quad Y_u = \frac{2}{3}, \quad Y_d = -\frac{1}{3}, \quad Y_L = -\frac{1}{2}, \quad Y_e = -1.$$

These are the unique hypercharges (up to normalization and $u \leftrightarrow d$) consistent with anomaly cancellation for the derived template with $N_c = 3$.

Consequences. Electric charges $Q = T_3 + Y$ give $Q(u) = 2/3$, $Q(d) = -1/3$, $Q(e) = -1$, $Q(\nu) = 0$. Charge *ratios* are fixed by the anomaly structure; with the compact- $U(1)$ unit convention $Y_Q = 1/6$, all electric charges land on multiples of $1/3$. The ratios are derived; the unit is the convention (§7). The quark–lepton relation $Y_L = -N_c Y_Q$ ties lepton hypercharge to the number of colors.¹⁰

¹⁰Code reference: `check_T_field` (hypercharge section) in `gauge.py`.

7. The one-U(1) a-posteriori closure theorems

The abelian sector’s uniqueness results are stated here as *a-posteriori closure theorems* for the selected template. The logical order is: gauge template (consumed at C2 of Theorem 5.9) $\rightarrow T_{\text{field}}$ (the scan, §5) $\rightarrow$ the closure theorems below $\rightarrow L_{\text{hypercharge}}$ (§6). Proposition 7.1 is template-independent: for *any* fixed set of non-abelian multiplets $\mathcal{M}$ satisfying EC and CM, the number of admissible U(1) factors is determined by the multiplet separation problem. Theorem 7.7 is its SM specialization.

There is no circularity, and the claim structure is worth stating plainly. The scan takes the template — with one U(1) available to grade — as an input: F6 solves hypercharge assignments on that template, so the scan consumes the abelian slot, not only $\text{SU}(3) \times \text{SU}(2)$ quantum numbers. The closure theorems then verify, for the scanned matter content, that the abelian sector is consistent and that its minimal labeling is unique: at least one abelian grading is necessary (under R-EC-inv + I-typing), one suffices, and no second is admissible under CM. Neither result claims to derive the other’s input: the scan does not derive the template, and the closure theorems do not derive the matter content. A mutual-consistency reading is available as the honest strong form — the (template, matter content, single-abelian-grading) triple is a simultaneous solution of the full constraint set — and whether it is the *unique* simultaneous solution is the fixed-point program, the head of the open-lane register of Technical Supplement II (*The Foundational Gauge Program*).

The unique abelian factor result is split into two levels. **Theorem A** (Proposition 7.1) is template-independent: it shows that for *any* multiplet set $\mathcal{M}$ and *any* block decomposition, EC and CM determine the number of admissible abelian factors from the separation deficiency δ alone. **Theorem B** (Theorem 7.7) is the SM specialization: it plugs in $\{n_k\} = \{3, 2, 1\}$ and $\mathcal{M} = \mathcal{M}_{\text{SM}}$ and outputs “exactly one U(1), uniquely hypercharge.”

Proposition 7.1 (Unique minimal abelian grading factor — structural (Theorem A)). *Let $\mathcal{A} \cong \bigoplus_{\alpha=1}^m M_{n_\alpha}(\mathbb{C})$ be the admissibility algebra at a fixed interface, with non-abelian gauge factor $G_{\text{na}} = \prod_{\alpha=1}^m \text{SU}(n_\alpha)$ and relative-phase torus $T_{\text{rel}} := Z(\text{U}(\mathcal{A}))/\text{U}(1)_{\text{diag}} \cong \text{U}(1)^{m-1}$. Let $\mathcal{M} = \{r_1, \dots, r_N\}$ denote the set of physically distinct matter multiplet types occurring in the derived template, regarded as irreducible carriers of G_{na} .*

Assume that G_{phys} is an admissible gauge structure (Definition 2.20 of Technical Supplement II (The Foundational Gauge Program); EC and CM hold by Lemma 2.24 of Technical Supplement II (The Foundational Gauge Program)).

Lemma 7.2 (Abelian factors arise only from T_{rel}). *Every continuous abelian gauge factor of G_{phys} is a one-dimensional subgroup of T_{rel} . No abelian gauge factor can arise from any other source.*

Proof. By the exact sequence (Proposition 2.19 of Technical Supplement II (*The Foundational Gauge Program*)):

$$1 \rightarrow Z(\text{U}(\mathcal{A})) \rightarrow \text{U}(\mathcal{A}) \xrightarrow{\pi} \text{Inn}(\mathcal{A}) \rightarrow 1.$$

$\text{Inn}(\mathcal{A}) = \prod_{\alpha} \text{PU}(n_{\alpha})$ is semisimple (each $\text{PU}(n)$ is semisimple for $n \geq 2$, and trivial for $n = 1$): its Lie algebra has no nonzero abelian *ideal*, equivalently the group has no connected abelian *normal direct factor*. (It certainly contains continuous abelian subgroups — every maximal torus is one; what semisimplicity excludes is an abelian factor that could serve as an independent gauge direction with its own coupling.) The abelian continuous subgroups of $\text{U}(\mathcal{A}) = \prod_{\alpha} \text{U}(n_{\alpha})$ that commute with the whole group — the candidates for an independent abelian gauge factor — are contained in $\ker(\pi) = Z(\text{U}(\mathcal{A}))$. Quotienting by the physically unobservable diagonal $\text{U}(1)_{\text{diag}}$ gives $T_{\text{rel}} = Z/\text{U}(1)_{\text{diag}}$ as the full group of continuous abelian gauge transformations. Any abelian gauge factor is therefore a one-dimensional subgroup of T_{rel} .

The argument excludes the following loopholes:

- (a) *Abelian subgroups of $\text{Inn}(\mathcal{A})$?* $\text{PU}(n)$ has a maximal torus of dimension $n - 1$ — an abelian subgroup, but not an abelian *normal factor*: its elements are non-central for $n \geq 2$, it is conjugated non-trivially, and its generators are already counted in G_{na} as Cartan directions with the non-abelian factor's single coupling. A torus inside a simple factor is not an independent abelian gauge direction; only a normal abelian direct factor would be, and semisimplicity excludes exactly that.
- (b) *Discrete abelian factors?* Discrete subgroups (e.g., $\mathbb{Z}_N$) can survive as global symmetries but do not contribute continuous gauge generators. CM counts continuous admissibility channels; discrete symmetries have no per-generator capacity cost.
- (c) *External abelian factors?* Any abelian factor not contained in $\text{U}(\mathcal{A})$ would act non-trivially on $\mathcal{H}_\omega$ but not arise from the admissibility algebra. This contradicts the gauge identification theorem (Theorem 2.11 of Technical Supplement II (*The Foundational Gauge Program*)): the gauge group is $\text{Inn}(\mathcal{A})$ (plus the central extensions from the exact sequence), and no external group acts.

□

Definition 7.3 (Separation deficiency). The *separation deficiency* of G_{na} on $\mathcal{M}$ is the number of pairs of physically distinct multiplet types that share identical non-abelian quantum numbers:

$$\delta(G_{\text{na}}, \mathcal{M}) := |\{(r_i, r_j) : i < j, [r_i]_{G_{\text{na}}} = [r_j]_{G_{\text{na}}}\}|.$$

EC is satisfied by G_{na} alone iff $\delta = 0$. If $\delta > 0$, additional abelian grading is required to resolve the remaining degeneracies.

Lemma 7.4 (Minimum number of abelian factors from separation deficiency). *Let $\mathcal{P} = \{P_1, \dots, P_s\}$ be the partition of $\mathcal{M}$ into equivalence classes under the non-abelian label map (i.e., $P_k = \{r_i : [r_i]_{G_{\text{na}}} = \ell_k\}$ for each distinct non-abelian label ℓ_k). Let $m_{\text{max}} := \max_k |P_k|$ be the size of the largest equivalence class. Then:*

- (i) *At least $\lceil \log_2 m_{\text{max}} \rceil$ binary distinctions are needed to separate the elements of the largest class.*
- (ii) *A single $\text{U}(1)$ factor assigns a real-valued charge Y_i to each $r_i \in P_k$. If the Y_i are pairwise distinct within each class P_k , then one $\text{U}(1)$ suffices to separate all multiplets.*
- (iii) *Therefore, the minimum number of abelian factors is 0 if $\delta = 0$, and 1 if $\delta > 0$ and there exists a charge assignment Y that separates all classes.*

Proof. (i) is standard: m_{max} elements require $\lceil \log_2 m_{\text{max}} \rceil$ binary attributes to distinguish. (ii): a single continuous $\text{U}(1)$ charge $Y \in \mathbb{R}$ has cardinality $\mathfrak{c}$, so it can separate any finite number of elements — the condition is that the charge assignment $Y : P_k \rightarrow \mathbb{R}$ is injective on each class. (iii) combines (i) and (ii): if $m_{\text{max}} \geq 2$, at least one abelian factor is needed; if an injective Y exists, one suffices. □

Lemma 7.5 (Diagonal rational grading from Schur + anomaly cancellation). *Each $\text{U}(1) \subset T_{\text{rel}}$ assigns a rational charge to each r_i , constant on that multiplet type. Both properties are derived:*

- (a) **Diagonal action (Schur).** *Each r_i is an irreducible representation of $G_{\text{na}} = \prod_\alpha \text{SU}(n_\alpha)$. The relative-phase torus $T_{\text{rel}} \subset Z(\text{U}(\mathcal{A}))$ lies in the center of the unitary group. By Schur's lemma, any central element acts on an irreducible representation by a scalar. Therefore each $\text{U}(1) \subset T_{\text{rel}}$ acts on r_i by a phase $e^{iY_i\theta}$, where $Y_i \in \mathbb{R}$ depends only on the multiplet type, not on the state within the multiplet. This is diagonal action.*

(b) **Rational charges (anomaly cancellation).** The charges Y_i are constrained by gauge anomaly cancellation (§6): the four conditions $[\text{SU}(2)]^2[\text{U}(1)]$, $[\text{SU}(3)]^2[\text{U}(1)]$, $[\text{grav}]^2[\text{U}(1)]$, $[\text{U}(1)]^3$ are polynomial equations in $\{Y_i\}$ with integer coefficients. Solving the first three linearly and substituting into the fourth yields (Theorem 6.1): $z^2 - 2z - (N_c^2 - 1) = 0$, where $z := Y_u/Y_Q$. This polynomial is monic (leading coefficient = 1) with integer coefficients. By the rational root theorem for monic integer polynomials, any rational root p/q (in lowest terms) must have $q \mid 1$, i.e., $z \in \mathbb{Z}$. Indeed, $z = 1 \pm N_c$ are integers. Since all other Y_i are rational multiples of Y_Q (from the linear conditions), and the standard normalization $Y_Q = 1/6$ (a convention: the anomaly system is homogeneous and determines only the ray of charges, up to the discrete $u \leftrightarrow d$ exchange), all charge ratios are rational. What the anomaly system proves is rationality of ratios; the absolute charge unit is fixed by the compactness of the $\text{U}(1)$ and the choice of character lattice made in the global-group ($\mathbb{Z}_6$ -quotient) analysis, and choosing $Y_Q = 1/6$ is that convention, not a derivation of rationality from anomalies alone (the compact- $\text{U}(1)$ representation-data premise; front-matter imported-ingredient ledger).

Both properties were previously listed as assumption (A3). The derivation shows they follow from Schur’s lemma (standard representation theory, no additional physical input) and anomaly cancellation (already derived in this supplement). No assumption beyond EC, CM, the selected gauge template, and the scanned matter content is needed.

Then:

- (i) If $\delta(G_{\text{na}}, \mathcal{M}) > 0$, then at least one nontrivial $\text{U}(1) \subset T_{\text{rel}}$ is necessary (from EC: unresolved degeneracies violate admissibility completeness).
- (ii) If there exists a charge assignment $Y : \mathcal{M} \rightarrow \mathbb{Q}$ such that the augmented label map $r_i \mapsto ([r_i]_{G_{\text{na}}}, Y(r_i))$ is injective, then one abelian grading factor is sufficient (Lemma 7.4(ii)).
- (iii) Any additional independent abelian factor Y' is inadmissible unless it changes the induced partition of $\mathcal{M}$; if Y already separates all types, every further $\text{U}(1)$ is redundant and excluded by CM (Lemma 2.22 of Technical Supplement II (The Foundational Gauge Program): each redundant generator wastes ε^* capacity).
- (iv) Therefore, whenever $\delta > 0$ and one abelian grading separates all types, there exists exactly one physically admissible abelian gauge factor, unique up to overall normalization and change of basis in T_{rel} .

Scope. Uniqueness here is uniqueness relative to the EC/CM criterion: the claim is that once one abelian grading separates all multiplet types, any further independent $\text{U}(1)$ current adds a generator (cost $\geq \varepsilon^*$, Lemma 2.22 of Technical Supplement II (The Foundational Gauge Program)) while changing no gauge label assignment, and A2 eliminates it. The criterion by which a second $\text{U}(1)$ could earn its keep is label separation; a hypothetical second current that is “physically independent” in some respect not expressible as a label distinction is outside EC’s vocabulary and is excluded by CM’s cost accounting, not by a separation argument. This is a premise-level commitment of the admissibility semantics (Definition 2.20 of Technical Supplement II (The Foundational Gauge Program)), named here so the reader can see exactly what carries the exclusion. One further clause is owned in the same register: a second abelian current distinct from hypercharge only dynamically — identical kinematic labels on every multiplet, different couplings or breaking pattern — is excluded by the EC/CM vocabulary itself (kinematic type-distinctness exhausted by counted labels, R-EC-inv), not by finite capacity; dynamically distinguished currents are outside the reading’s scope by construction, and admitting them would change the count.

Proof. Step 1: Necessity (at least one abelian factor). If $\delta(G_{\text{na}}, \mathcal{M}) > 0$, there exist distinct

multiplet types $r_i \neq r_j$ with $[r_i]_{G_{\text{na}}} = [r_j]_{G_{\text{na}}}$. By EC (Lemma 2.24 of Technical Supplement II (*The Foundational Gauge Program*)), these two physical types must be distinguishable by the gauge structure. Since G_{na} does not separate them, the distinction must come from the remaining gauge freedom. By Lemma 7.2, the only continuous abelian gauge freedom is T_{rel} . Hence at least one nontrivial $U(1) \subset T_{\text{rel}}$ is necessary. This gives (i).

Step 2: Sufficiency of one factor (explicit separation matrix). We construct the separation matrix S for $\mathcal{M}_{\text{SM}}$. The non-abelian equivalence classes are:

Class	Non-abelian label	Members
P_1	$(3, 2)$	$\{Q\}$ (singleton)
P_2	$(\bar{3}, 1)$	$\{u^c, d^c\}$
P_3	$(1, 2)$	$\{L\}$ (singleton)
P_4	$(1, 1)$	$\{e^c\}$ (singleton)

The separation deficiency is $\delta = |P_2|(|P_2| - 1)/2 = 1$: a single unresolved pair (u^c, d^c) . The maximum class size is $m_{\text{max}} = 2$. By Lemma 7.4, one $U(1)$ factor suffices iff it assigns distinct charges to u^c and d^c .

Chirality/sign convention (fixed corpus-wide in v4.0). Template entries are *left-handed Weyl fields*: u^c, d^c, e^c name the LH conjugates in representations $(\bar{\mathbf{3}}, \mathbf{1})$, $(\bar{\mathbf{3}}, \mathbf{1})$, $(\mathbf{1}, \mathbf{1})$, with hypercharges $Y(u^c) = -2/3$, $Y(d^c) = +1/3$, $Y(e^c) = +1$. Numerical charge *values* in this document are quoted for the right-handed partner fields, $Y_u := Y(u_R) = -Y(u^c) = +2/3$, $Y_d := Y(d_R) = -1/3$, $Y_e := Y(e_R) = -1$, because those are the conventional tabulated values. Representation conjugation and $U(1)$ sign always flip together; v3.9 paired the c symbols with the right-handed values, which mixed the two conventions — the notation below now uses Y_u, Y_d, Y_e (right-handed values) throughout, alongside the LH template reps. The anomaly system is invariant under the joint flip, so no computed quantity changes.

The hypercharge assignment $Y_u = 2/3 \neq -1/3 = Y_d$ resolves this degeneracy. The full charge vector $(Y_Q, Y_u, Y_d, Y_L, Y_e) = (1/6, 2/3, -1/3, -1/2, -1)$ is pairwise distinct across all five multiplet types. The augmented label map

$$r_i \mapsto ([r_i]_{\text{SU}(3) \times \text{SU}(2)}, Y(r_i))$$

is therefore injective on $\mathcal{M}_{\text{SM}}$: no two multiplets share both their non-abelian quantum numbers and their hypercharge. One $U(1)$ suffices. This gives (ii).

Step 3: No second abelian factor needed or admissible. Suppose a second independent abelian factor with charge assignment $Y' : \mathcal{M} \rightarrow \mathbb{Q}$ is added. The augmented map becomes $r_i \mapsto ([r_i]_{G_{\text{na}}}, Y(r_i), Y'(r_i))$. Since Y already makes this map injective (Step 2), the additional charge Y' does not refine the partition of $\mathcal{M}$: the partition $\{\{Q\}, \{u^c\}, \{d^c\}, \{L\}, \{e^c\}\}$ is already the finest possible (each class is a singleton). By CM (Lemma 2.24 of Technical Supplement II (*The Foundational Gauge Program*)), the generator of Y' consumes $\varepsilon^* > 0$ capacity (Lemma 2.22 of Technical Supplement II (*The Foundational Gauge Program*)) with zero admissibility gain. Under finite capacity (A1), this is inadmissible. This gives (iii).

Step 4: Uniqueness up to normalization. The surviving abelian factor is a one-dimensional subgroup of $T_{\text{rel}} \cong U(1)^{m-1}$. In the SM case, $m = 3$ (blocks M_3, M_2, M_1), so $T_{\text{rel}} \cong U(1)^2$. The requirement that the charge assignment resolves the (u^c, d^c) degeneracy and satisfies all four anomaly conditions (§6) determines a *unique direction* in T_{rel} , up to normalization ($Y \mapsto \lambda Y$, $\lambda \in \mathbb{Q}^\times$). This uniqueness

is proved in Theorem 6.1: the anomaly system has exactly two solutions $z = 1 \pm N_c$ (with $N_c = 3$: $z = 4$ and $z = -2$), which correspond to the same direction under the $u \leftrightarrow d$ relabeling symmetry. Therefore the abelian factor is unique. This gives (iv). $\square$

Corollary 7.6 (Application to the derived Standard Model template). *For the derived multiplet set $\mathcal{M}_{\text{SM}} = \{Q, u^c, d^c, L, e^c\}$, the non-abelian labels under $\text{SU}(3) \times \text{SU}(2)$ are $Q \sim (3, 2)$, $L \sim (1, 2)$, $u^c \sim (\bar{3}, 1)$, $d^c \sim (\bar{3}, 1)$, $e^c \sim (1, 1)$. Hence the non-abelian label map is not injective (it conflates u^c with d^c): $\delta = 1$. Hypercharge resolves this degeneracy with pairwise distinct charges. By Proposition 7.1, exactly one abelian gauge factor is necessary and sufficient.*

The preceding chain of lemmas, proposition, and corollary assembles the result from reusable components. The reviewer asks for a single self-contained theorem that takes the exact-sequence data as input and outputs “exactly one $\text{U}(1)$ ” without requiring the reader to trace through the intermediate results. We provide this.

Theorem 7.7 (Unique abelian factor from exact-sequence data — SM specialization (Theorem B)).
Hypotheses:

- (H1) **Admissibility algebra.** $\mathcal{A} = M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}$ (Wedderburn blocks $\{n_k\} = \{3, 2, 1\}$, from T2a + Theorem R).
- (H2) **Exact sequence.** $1 \rightarrow Z(\text{U}(\mathcal{A})) \rightarrow \text{U}(\mathcal{A}) \xrightarrow{\pi} \text{Inn}(\mathcal{A}) \rightarrow 1$, with $\text{Inn}(\mathcal{A}) = \text{PU}(3) \times \text{PU}(2)$, $\ker(\pi) = \text{U}(1)^3$, $T_{\text{rel}} = \ker(\pi)/\text{U}(1)_{\text{diag}} \cong \text{U}(1)^2$ (Proposition 2.19 of Technical Supplement II (The Foundational Gauge Program)).
- (H3) **Matter content.** $\mathcal{M} = \{Q(3, 2), u^c(\bar{3}, 1), d^c(\bar{3}, 1), L(1, 2), e^c(1, 1)\}$ (from T_{field} , §5). This hypothesis is the consumption site of the entry-typed inventory I-typing: the five listed multiplets are antecedently distinct types — $u^c \neq d^c$ prior to any abelian label is an input here, not a consequence of the labels — and Definition 2.20 of Technical Supplement II (The Foundational Gauge Program)’s copy clause is scoped to the full realized $\tilde{G}$ precisely so that this inventory is not recounted under restriction to proper subgroups.
- (H4) **EC and CM.** The gauge structure satisfies admissibility completeness (EC: the label map is injective) and capacity minimality (CM: no generator can be removed while preserving EC); EC holds under the kinematic type-inventory reading R-EC-inv with the entry-typed inventory I-typing (the Definition 2.20 of Technical Supplement II (The Foundational Gauge Program) scoping premise and Lemma 2.24 of Technical Supplement II (The Foundational Gauge Program)’s premise list); CM from PLEC’s A2 component over the A1 + MD-defined admissible set (Lemma 2.24 of Technical Supplement II (The Foundational Gauge Program)).

Conclusion: Under (H1)–(H4) — in particular under EC as read through R-EC-inv with I-typing (H3–H4) — the physical gauge group contains exactly one continuous abelian factor, which is a uniquely determined one-dimensional subgroup of $T_{\text{rel}} \cong \text{U}(1)^2$. This factor is hypercharge $\text{U}(1)_Y$, with charge assignments $(Y_Q, Y_u, Y_d, Y_L, Y_e) = (1/6, 2/3, -1/3, -1/2, -1)$ up to normalization (the compact- $\text{U}(1)$ charge-lattice convention; front-matter ledger) and the $u \leftrightarrow d$ relabeling.

Grades: the necessity leg (≥ 1 abelian factor) carries $[P_{\text{structural_reading}} \mid \text{R-EC-inv} + \text{I-typing}]$; the sufficiency and uniqueness-of-direction legs are unconditional given the anomaly system; the CM leg (no second factor) is $[P]$ (A2 over the A1 + MD admissible set).

Proof. The proof proceeds in five steps: source exclusivity, necessity, sufficiency, uniqueness of direction, and redundancy of any second factor.

Step 1 (Source exclusivity): every abelian factor lives in T_{rel} . By (H2), $\text{Inn}(\mathcal{A}) = \text{PU}(3) \times \text{PU}(2)$ is

semisimple (both factors are semisimple for $n \geq 2$). A semisimple group has no continuous abelian quotients: every connected abelian subgroup of $\mathrm{PU}(n)$ is a maximal torus (the image of the Cartan subalgebra of $\mathfrak{su}(n)$), which is part of the non-abelian structure. The only continuous abelian subgroups of $\mathrm{U}(\mathcal{A})$ that project to the identity under π are in $\ker(\pi) = Z(\mathrm{U}(\mathcal{A}))$. Quotienting by the unobservable $\mathrm{U}(1)_{\mathrm{diag}}$ gives $T_{\mathrm{rel}} \cong \mathrm{U}(1)^2$. Any abelian gauge factor is a one-dimensional subgroup of this torus.

Step 2 (Necessity): at least one $\mathrm{U}(1)$ is required. By (H3), the non-abelian label map $r_i \mapsto [r_i]_{\mathrm{SU}(3) \times \mathrm{SU}(2)}$ is:

r_i	Q	u^c	d^c	L	e^c
$[r_i]$	$(3, 2)$	$(\bar{3}, 1)$	$(\bar{3}, 1)$	$(1, 2)$	$(1, 1)$

The map sends both u^c and d^c to $(\bar{3}, 1)$. Therefore the label map is not injective: $\delta(\mathrm{SU}(3) \times \mathrm{SU}(2), \mathcal{M}) = 1$. By EC (H4), all five types must be distinguished by the gauge structure. Since the non-abelian labels fail to separate u^c from d^c , at least one abelian factor from T_{rel} is required.

Step 3 (Sufficiency): one $\mathrm{U}(1)$ resolves all degeneracies. A one-dimensional subgroup $\mathrm{U}(1) \subset T_{\mathrm{rel}}$ assigns a rational charge Y_i to each r_i (by Schur’s lemma: central elements act by scalars on irreducible representations; by the anomaly system: the charges are rational, §6). The augmented label map $r_i \mapsto ([r_i]_{\mathrm{SU}(3) \times \mathrm{SU}(2)}, Y_i)$ is injective iff $Y_u \neq Y_d$ (this is the only unresolved pair from Step 2; the other three types already have distinct non-abelian labels). The anomaly system (Theorem 6.1) determines:

$$Y_u = z \cdot Y_Q, \quad Y_d = (2 - z) \cdot Y_Q,$$

where z satisfies $z^2 - 2z - (N_c^2 - 1) = 0$ with $N_c = 3$, giving $z \in \{4, -2\}$. For either root: $Y_u - Y_d = (2z - 2)Y_Q \neq 0$ (since $z \neq 1$). Therefore one $\mathrm{U}(1)$ suffices.

Step 4 (Uniqueness of direction): the anomaly system admits a unique direction in T_{rel} . $T_{\mathrm{rel}} \cong \mathrm{U}(1)^2$ is a two-dimensional torus. A one-dimensional subgroup is specified by a direction $\mathbf{q} = (q_1, q_2) \in \mathbb{Z}^2 \setminus \{0\}$ (up to positive rescaling). The charge of multiplet r_i is $Y_i = q_1 \cdot c_1(r_i) + q_2 \cdot c_2(r_i)$, where c_1, c_2 are the canonical $\mathrm{U}(1)$ charges from the center $Z = \mathrm{U}(1)_3 \times \mathrm{U}(1)_2 \times \mathrm{U}(1)_1$. (In the canonical basis: $c_1(r_i) = n_3$ -ality of the $\mathrm{SU}(3)$ rep, $c_2(r_i) = \mathrm{SU}(2)$ isospin parity.)

The four anomaly conditions (§6) are four polynomial equations in (q_1, q_2) . Three are linear; the fourth is cubic. The linear conditions determine the charge ratios $Y_u/Y_Q, Y_d/Y_Q, Y_L/Y_Q, Y_e/Y_Q$ as functions of a single parameter $z = Y_u/Y_Q$. The cubic condition reduces to the quadratic $z^2 - 2z - 8 = 0$ with roots $z = 4$ and $z = -2$. These two roots correspond to the same direction in T_{rel} (they are related by the $u \leftrightarrow d$ relabeling symmetry of the fermion template). Therefore the anomaly system admits a *unique direction* $\mathbf{q}$ in T_{rel} , up to normalization and the discrete relabeling. (“Admits” rather than “selects”: no mechanism ranges over candidate directions; the admissibility conditions — four anomaly equations plus EC, CM — have a unique non-trivial common solution up to the stated equivalences, and that solution is the realised admissible hypercharge — admissibility space’s argmin (A2) on the Y-assignment alternatives.)

Step 5 (No second factor): CM excludes any additional $\mathrm{U}(1)$. After Step 3, the augmented label map $r_i \mapsto ([r_i]_{\mathrm{SU}(3) \times \mathrm{SU}(2)}, Y_i)$ is injective on all five multiplet types (the full charge vector $(1/6, 2/3, -1/3, -1/2, -1)$ is pairwise distinct). The partition of $\mathcal{M}$ into equivalence classes under this map is $\{\{Q\}, \{u^c\}, \{d^c\}, \{L\}, \{e^c\}\}$ — the discrete partition (all singletons).

A second independent $U(1)' \subset T_{\text{rel}}$ would assign a second charge Y_i' to each multiplet. The augmented map $r_i \mapsto ([r_i], Y_i, Y_i')$ induces the same discrete partition (singletons cannot be further refined). Therefore Y' provides zero additional admissibility information. By CM (H4), the generator of $U(1)'$ occupies one admissibility channel (Lemma 2.22 of Technical Supplement II (*The Foundational Gauge Program*)) at cost $\varepsilon^* > 0$ (L_{ε^*}) with zero admissibility gain. Under finite capacity (A1), this is inadmissible.

Conclusion. Steps 1–5 establish: exactly one continuous abelian factor exists in the physical gauge group; it is a one-dimensional subgroup of $T_{\text{rel}} \cong U(1)^2$; the anomaly system selects a unique direction; and no second factor is admissible. The physical gauge group is therefore $[SU(3) \times SU(2) \times U(1)_Y]/\mathbb{Z}_6$ (Proposition 2.26 of Technical Supplement II (*The Foundational Gauge Program*)). $\square$

Unique $U(1)$ proof in conventional language

Starting from $\mathcal{A} = M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}$, the nonabelian inner automorphism group contributes $PU(3) \times PU(2)$ and no central $U(1)$. The unitary-center quotient leaves a two-dimensional relative-phase torus. The derived matter template contains two multiplets with the same nonabelian label, u^c and d^c , so admissibility completeness requires at least one relative phase. The standard anomaly equations on the scanned template reduce that torus to one rational projective ray, hypercharge. Once this ray is included, all five multiplets are separated; any second independent ray either violates the anomaly equations on the 45-Weyl template or separates no additional minimal-template distinction. MD assigns that extra generator positive cost, so A2 removes it. This is the whole no-redundancy proof, stated without APF shorthand.

Why the theorem is not heuristic. The selection of one $U(1)$ from the $(K - 1)$ -torus involves three independently verifiable steps, none of which is an aesthetic or heuristic choice:

- *Necessity* (Step 2) is a counting argument: $\delta > 0$ combined with EC forces at least one abelian factor. This is a finite combinatorial check on the five multiplets.
- *Uniqueness of direction* (Step 4) is a number-theoretic result: the anomaly system is a determined system of polynomial equations with a unique projective solution. This is verified by explicit computation (the discriminant of $z^2 - 2z - 8 = 0$ is 36; the roots are 4 and -2 ; they are related by the relabeling symmetry).
- *Redundancy exclusion* (Step 5) is a capacity argument: the partition is already maximally fine (all singletons), so no new information can be added. This is a logical tautology combined with the generator–channel bridge (each generator costs $\varepsilon^* > 0$).

The only conditional input is the matter content $\mathcal{M} = \mathcal{M}_{\text{SM}}$ (H3), which is derived in T_{field} (§5). Given that input, the conclusion is a theorem, not a selection.

Edge case: $U(1)_{B-L}$ with sterile ν_R . A frequently raised alternative is the anomaly-free combination $U(1)_{B-L}$ (baryon minus lepton number), which becomes anomaly-free when the multiplet set is extended by three right-handed neutrinos $\nu_R(\mathbf{1}, \mathbf{1})$. In this enlarged template $\mathcal{M}' = \mathcal{M}_{\text{SM}} \cup \{\nu_R\}$, both $U(1)_Y$ and $U(1)_{B-L}$ are anomaly-free, and the question is whether two independent abelian factors are admissible.

The APF excludes this scenario at two stages:

- T_{field} **excludes ν_R from the minimal template.** The fermion scan (§5) derives $\mathcal{M}_{\text{SM}}$ as the unique minimum-DOF survivor at 45 Weyl fermions. The template $\mathcal{M}_{\text{SM}} \cup \{\nu_R\}$ has 48 DOF and survives all anomaly filters, but is non-minimal: it is the second-lightest post-CPT survivor (template #3 in the waterfall table). Since the multiplet set is $\mathcal{M}_{\text{SM}}$, not

$\mathcal{M}'$, the $B-L$ direction is not anomaly-free for the *derived* content. (Indeed, $B-L$ has a mixed gravitational anomaly $[\text{grav}]^2[\text{U}(1)_{B-L}] = 3 \neq 0$ unless ν_R is present.)

- (b) **Capacity minimality excludes the second $\text{U}(1)$ even if ν_R were present.** Suppose counterfactually that $\mathcal{M}' = \mathcal{M}_{\text{SM}} \cup \{\nu_R\}$ were the derived template. Does Proposition 7.1 still exclude a second $\text{U}(1)$? Yes: in $\mathcal{M}'$, the non-abelian labels conflate u^c/d^c (both $(\bar{3}, 1)$) and e^c/ν_R (both $(1, 1)$). A single abelian grading with four distinct charges resolves all degeneracies: one $\text{U}(1)$ suffices. $\text{U}(1)_{B-L}$ is then a *linear combination* of this grading and the non-abelian quantum numbers—not an independent gauge direction. A second independent $\text{U}(1)$ would carry capacity cost (one additional admissibility channel) with no additional multiplet-separation power, violating CM.
- (c) **Observational cross-check (bridge-conditional, exited).** Under the cosmological identification (BA6, §3) the ν_R -extended count $C_{\text{total}} = 64$ reads $\Omega_\Lambda = 42/64$ against the 45-template’s 42/61; the comparison with the Planck measurement lives at the corpus homes (Papers 8 and 41, with the exited cosmology tail). The core exclusion above is by minimality and consumes no bridge axiom.

In summary, the “exactly one $\text{U}(1)$ ” conclusion is robust against the $B-L$ edge case: the extended multiplet set is non-minimal (T_{field}), $B-L$ is anomalous for the minimal template, and even if ν_R were included, $B-L$ would be a redundant linear combination rather than an independent gauge factor.

String-theoretic extra $\text{U}(1)$ ’s and Stückelberg masses. Heterotic string compactifications and F-theory models generically produce multiple abelian gauge factors beyond hypercharge. In these constructions, unwanted $\text{U}(1)$ ’s acquire large masses via the Stückelberg mechanism: the $\text{U}(1)$ gauge boson “eats” a closed-string axion, becoming massive at or near the compactification scale, while the associated global selection rule (e.g., a $\mathbb{Z}_N$ discrete symmetry) may persist at low energies.

In the APF, this scenario has a precise capacity interpretation:

- (a) A Stückelberg-massive $\text{U}(1)$ is a gauge boson that no longer propagates as a long-range force. In the admissibility language, the corresponding channel is “saturated”: its admissibility cost is paid at the compactification scale, and at low energies it is absorbed into the vacuum sector (L_{vac} ; exited to Paper 45 at the v4.0 core split, BA9 in §3). The channel is counted in C_{total} (as part of the UV structure) but does not contribute an independent low-energy gauge coupling.
- (b) Capacity minimality (CM) does not merely select the gauge group with the fewest generators; it selects the group with the fewest *active admissibility channels* at the IR saturation scale. A Stückelberg-massive $\text{U}(1)$ is inactive at low energies and therefore excluded from the low-energy gauge group by the same principle that excludes ν_R from the low-energy fermion count: it is above the IR capacity ledger.
- (c) The APF is therefore selecting the *phase* in which only one $\text{U}(1)$ combination remains both anomaly-free and massless at low energies. This is precisely the phase that heterotic phenomenology identifies as the “Standard Model embedding”: among the multiple $\text{U}(1)$ ’s of the string construction, one linear combination is identified with hypercharge and remains massless, while the others acquire Stückelberg masses and decouple. The APF’s capacity-minimality principle provides the *selection criterion* for this phase: the massless $\text{U}(1)$ is the one that resolves all low-energy multiplet degeneracies at minimum capacity cost.

Kinetic mixing. Kinetic mixing — a gauge-invariant term $\frac{\chi}{2} F_{\mu\nu} F'^{\mu\nu}$ coupling two abelian field strengths — needs a second propagating $\text{U}(1)$, and Proposition 7.1 leaves none to mix with. The concern is moot for the low-energy gauge structure derived here, and we say no more about it in the

classification core: the capacity accounting of hypothetical mixed pairs is program-register material and belongs with the foundational program, not among the closure theorems.

Roadmap: from PU automorphisms to the physical SM group

The gauge derivation has four logically distinct layers.

1. **Algebra layer:** inner automorphisms of the finite semisimple admissibility algebra give $\prod_k \text{PU}(n_k)$. This is the adjoint action on observables.
2. **Field layer:** matter multiplets are vectors/modules in $\mathcal{H}_\omega$, so projective $\text{PU}(n)$ actions must be lifted to linear $\text{SU}(n)$ actions whenever fundamental matter exists.
3. **Abelian layer:** $\text{U}(1)_Y$ does not come from $\text{Inn}(\mathcal{A})$; it comes from the relative-phase torus $Z(\text{U}(\mathcal{A}))/\text{U}(1)_{\text{diag}}$, pruned by the no-redundancy theorem to a single anomaly-compatible direction.
4. **Global layer:** central elements acting trivially on all derived matter multiplets are quotiented. For the SM hypercharge ray this kernel is $\mathbb{Z}_6$.

This box is the local answer to the common ambiguity “why $\text{Inn}(\mathcal{A}) = \prod \text{PU}(n)$ but the physical group is $[\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)_Y]/\mathbb{Z}_6$.” The first group acts faithfully on observables; the second acts faithfully on the derived matter content.

Theorem 7.8 (No-redundancy theorem for extra abelian factors). *In the minimal APF gauge/matter theorem, the only continuous abelian gauge factor is the hypercharge ray $\text{U}(1)_Y$. Any additional continuous abelian factor is excluded by one of three mutually exclusive mechanisms:*

- (i) *it is a redundant central direction whose charges are an affine function of hypercharge on the derived template and therefore has strictly positive cost but no additional separating power;*
- (ii) *it is anomaly-incompatible on the 45-Weyl minimal template; or*
- (iii) *it separates distinctions not present in the minimal template, such as sterile-neutrino number or generation-resolved lepton number, and therefore belongs to a different enlarged admissibility problem.*

Consequently the theorem proves uniqueness of a single rational ray, not merely existence of one convenient $\text{U}(1)$ basis.

Proof. By Lemma 7.2, every continuous abelian gauge factor must come from the relative-phase torus $T_{\text{rel}} = Z(\text{U}(\mathcal{A}))/\text{U}(1)_{\text{diag}}$. Let q be any second abelian charge direction linearly independent of the hypercharge assignment Y on the minimal matter template.

If q is constant on each non-abelian degeneracy class already separated by Y , then replacing (Y, q) by Y preserves admissibility completeness. Since MD assigns a positive floor to each independent generator, A2 selects the one-generator realization. This is the capacity form of no redundancy.

If q acts nontrivially on the minimal 45-Weyl template, it must also satisfy the four anomaly equations. The anomaly system in Theorem 6.1, with the scanned representation content held fixed and no sterile singlet added, has a one-dimensional solution space; after minimal integral normalization this solution is precisely the hypercharge ray. Thus an independent q is not anomaly-compatible on that template.

The remaining familiar anomaly-free directions evade this conclusion only by changing the template. A $B - L$ direction becomes anomaly-compatible when a right-handed neutrino or equivalent sterile record is included. Family non-universal directions such as $L_\mu - L_\tau$ require generation labels to be independently gauge-addressable distinctions. Both moves add new primitive records, so they are outside the minimal theorem rather than counterexamples to it. The APF claim is therefore conditional and sharp: within the finite minimal template fixed here, exactly one abelian ray survives;

enlarged record vocabularies define enlarged theorems with additional cost.

The normalization is fixed as follows. The anomaly equations determine a ray in T_{rel} . Minimal charge quantization fixes the primitive integer representative by choosing $Y_{\text{int}} = 6Y$ with $\gcd\{|Y_{\text{int}}(r)| : r \in \mathcal{M}\} = 1$. Hence the continuous freedom is only the conventional gauge-coupling normalization, not an additional abelian factor. $\square$

Remark 7.9 (Right-handed neutrinos and enlarged abelian problems). If a sterile right-handed neutrino ν_R is included as a primitive matter record, the theorem's input template changes from the minimal 45-Weyl problem to a 48-Weyl problem. In that enlarged problem the standard anomaly equations admit an additional $B - L$ direction. APF therefore does not claim that $B - L$ is algebraically impossible. The claim is narrower: $B - L$ is not forced by the minimal gauge/matter template and it adds an independently admissible sterile or lepton-number distinction at positive MD cost. Hypercharge quantization on the minimal template remains a unique rational ray; in the ν_R -extended template the abelian solution space is larger, and A2 must be re-run on the enlarged admissible set rather than imported from the 45-Weyl theorem.

8. Scope of the three-generation input

This supplement proves the one-generation gauge/matter template and then uses the three-copy capacity ledger when reporting the 45-Weyl-fermion result. The necessity of exactly three generations is not re-derived here from first principles. It is imported from the capacity-counting branch of the program, and the import has named theorems, each machine-verified. The generation count is over-determined by three independent capacity-beta identities, each a linear equation with the unique solution $n = 3$: $6|b_3| = C_{\text{vacuum}} (66 - 8n = 9n + 15)$, $6|b_2| = C_{\text{matter}} (43 - 8n = 6n + 1)$ ¹¹, and the abelian member $6|b_Y| = d_{\text{eff}} - C_{\text{total}} ((40/3)n + 1 = 9n + 14)$ ¹². Because these are equations rather than one-sided bounds, they carry both directions: $n = 1, 2$, and 4 fail all three. Independently, the max-saturated selection at the electroweak wall kills under-replicated rivals outright ($n \in \{1, 2\}$ subsaturated, $n \geq 4$ over the ceiling)¹³. For the present supplement the statement is therefore conditional but explicit: if the capacity ledger supplies three copies of the one-generation template, then the fermion count is $3 \times 15 = 45$ Weyl fermions and the anomaly/hypercharge closure repeats generationwise. If one treats the generation number as a free external input, the gauge/hypercharge derivation still holds per generation, while the 45-count and cosmological denominator become bridge claims rather than outputs of this supplement alone.

9. Neutrino content of the minimum template

The supported statement of this section is the scan's own: *the minimum template in the declared scan contains no light sterile singlet*. The 45-fermion survivor carries no right-handed neutrino ν_R ; the nearest survivor that does (template #3 of the waterfall listing, at 48 DOF) sits strictly above the minimum and is excluded at the minimality cut. That is a classification fact about the declared scan space, and it is all this section claims.

What the absence of ν_R means for neutrino *masses* — the dimension-5 Weinberg-operator mechanism, the Majorana character of the resulting masses, and the neutrinoless-double-beta-decay observable — is physics downstream of the classification, stated at prediction register at the corpus homes (Papers 8 and 41, with the exited cosmology tail). So is the bridge-conditional observational argument against a light ν_R (the $C_{\text{total}} = 64$ reading under BA6). Neither is consumed by any theorem of this document.

Why ν_R is not counted. A heavy UV completion of neutrino mass (a Majorana state at a high scale Λ) is above the IR saturation scale and is not a distinct admissibility channel at low energies. The capacity count ($C_{\text{total}} = 61$) reflects the *low-energy* admissibility ledger; states above the saturation scale contribute to the UV completion but not to the structural partition.

Theorem 9.1 (ν_R extension theorem). *If a right-handed neutrino $\nu_R(\mathbf{1}, \mathbf{1})$ is added as a primitive matter distinction, the anomaly equations enlarge from the 45-Weyl minimal template to a 48-Weyl extended template. In that extended template, $B - L$ becomes anomaly-free, but it is not an additional abelian factor in the minimal APF sense unless the theory also pays for an independent sterile/generation-resolving admissibility record. A single abelian grading still separates all non-abelian degeneracies; a second massless $U(1)$ is capacity-redundant unless it carries new primitive record content. Thus ν_R is admissible as a non-minimal extension, but including it changes the*

¹¹Code reference: `check_L_beta_capacity` in `generations.py`.

¹²Code reference: `check_T_gauge_beta_capacity_tiling_abelian` in `gauge_beta_capacity_tiling.py`.

¹³Code reference: `check_R_Ngen_neq_3_killed` in `killed_rivals.py`.

theorem from “the unique minimum is 45 Weyl DOF with one $U(1)_Y$ ” to “a 48-Weyl extension admits additional anomaly-free linear combinations only when extra record distinctions are supplied.”

Proof. Adding ν_R cancels the mixed gravitational anomaly of $B - L$ and opens a larger abelian solution space in the standard anomaly equations. However, APF counts not only anomaly freedom but admissibility gain. Hypercharge already separates the non-abelian degeneracies u^c/d^c and e^c/ν_R in the extended set. A second abelian factor therefore distinguishes no additional minimal-template multiplet unless one declares the sterile or family label to be a new primitive record. Such a declaration is consistent, but it is an extension of the carrier and raises the ledger from the 45-Weyl theorem to a 48-Weyl theorem. The minimal result is therefore unaffected, while the extension is explicitly classified. $\square$

Where the physics readings live. The Majorana/ $0\nu\beta\beta$ prediction register and the bridge-conditional Planck comparison of the ν_R -extended count ($C_{\text{total}} = 64$ under BA6) are stated at the corpus homes (Papers 8 and 41, with the exited cosmology tail), where they carry their bridge conditioning. What stays here is the scan fact: no light sterile singlet in the minimum template. ¹⁴

¹⁴Code reference: `check_T_field` in `gauge.py`.

Part III

Capacity Counting

10. Capacity counting and the BRST question

10.1 Operational definition of capacity types

A *capacity type* is a structurally distinct admissibility role in the low-energy ledger. The count is:

$$C_{\text{total}} = \underbrace{45}_{\text{fermion types}} + \underbrace{4}_{\text{Higgs real components}} + \underbrace{12}_{\text{gauge generators}} = 61. \quad (13)$$

The count is stated under an explicit hypothesis: the channel-identification convention (BA5, §3; front-matter cost-uniqueness ledger row) that each Weyl fermion component, each real Higgs component, and each gauge generator occupies exactly one capacity slot of equal weight — the channel unit fixed by the cost functional ($L_{\text{Cauchy}}: F(n) = n$). The 61 is a theorem in that unit; sector-dependent slot weights would dissolve it (BA5’s stated failure mode).

Fermion types (45): the 15 Weyl fermions per generation ($Q = 6$ colored-doublet components + $u^c = 3$ colored-singlet + $d^c = 3$ + $L = 2$ weak-doublet components + $e^c = 1$) times 3 generations. Color and weak-isospin multiplicity *are* counted here as independent matter-sector admissibility roles, because each distinct quantum-number configuration is a separately admissible distinction in the capacity ledger. The arithmetic is $(6 + 3 + 3 + 2 + 1) \times 3 = 15 \times 3 = 45$, consistent with `check_L_count` in `gauge.py` (`per_gen = 6+3+3+2+1 = 15`). Internal-structure gauge channels (the 27 fermion-internal channels counted in L_{vac} , Paper 2 §7.1) are a distinct bookkeeping allocated to the vacuum sector; they are not the matter-sector fermion types counted here. This resolves an earlier prose formulation that read “the 5 multiplet types” and counted $5 \times 3 = 15$ rather than 45; the correct count is $(6 + 3 + 3 + 2 + 1) \times 3 = 45$.

Higgs components (4): the 4 real degrees of freedom of the Higgs doublet. After SSB, 3 become Goldstone bosons (absorbed into $W^\pm, Z$) and 1 is the physical Higgs. All 4 are counted because the capacity ledger operates at the structural level, before SSB selects a vacuum.

Gauge generators (12): $\dim(\text{SU}(3)) + \dim(\text{SU}(2)) + \dim(\text{U}(1)) = 8 + 3 + 1$. Each generator is a structurally distinct admissibility channel.

10.2 Why gauge constraints are not subtracted (the BRST question)

In perturbative quantum field theory, the physical Hilbert space is the BRST cohomology: gauge constraints subtract unphysical polarizations, reducing the propagating degrees of freedom. A natural question is: why does the capacity count not perform a similar subtraction?

The answer is that capacity types count *admissibility roles*, not *propagating degrees of freedom*. The distinction is:

Propagating DOF (QFT accounting): count on-shell polarization states. A massless gauge boson has 2 polarizations; BRST subtracts the longitudinal and timelike modes. This gives $2 \times 12 = 24$ propagating gauge DOF, not 12.

Enforcement roles (capacity accounting): count structurally distinct channels in the admissibility ledger. Each gauge generator $T^a \in \mathfrak{g}$ is one independent admissibility channel: it acts as a derivation on the algebra of observables, routing admissibility between matter fields. The number of generators is $\dim(\mathfrak{g}) = 12$, regardless of how many polarization states each generator supports.

The 12 gauge generators are assigned to the *vacuum* sector ($Q1 = 0$ in P_{exhaust}) precisely because they are admissibility *infrastructure*, not admissibility *consumers*. A gauge generator cannot route through itself; it has no upstream pathway to enter as a matter excitation. BRST is irrelevant to this accounting: it subtracts polarizations from the propagator, which is a kinematic object; the capacity count is a structural object.

Toy U(1) gauge model: constraint subtraction vs. channel accounting

A free Maxwell field has four components A_μ . Gauge fixing and BRST cohomology remove the longitudinal and timelike excitations, leaving two propagating photon polarizations. The APF ledger does not count four minus two; it counts one gauge-routing generator, because there is one independent local redundancy direction acting on charged matter by $\psi \mapsto e^{i\alpha}\psi$. For a nonabelian group the same rule gives one admissibility channel per Lie-algebra generator: $\dim \mathfrak{g}$. Ghosts and gauge-fixing fields are bookkeeping devices for quantizing the quotient; they do not supply new charged matter distinctions and do not erase the existence of the generator that must be enforced. This is why SU(3) contributes 8 structural channels, SU(2) contributes 3, and hypercharge contributes 1, even though the propagating polarization count is different.

15

10.3 I2 (gauge-cosmological identity): the $K = 61$ shared integer — exit note

What this supplement owns is the integer $C_{\text{total}} = 61$ and its structural partition $61 = 45 + 4 + 12$ (L_{count} , above). The identity I2 — that the same integer reappears as the common denominator of the cosmological projection, with the residual partition $3 + 16 + 42$ — is stated, proved against the ACC record, and compared with observation at its corpus home (Paper 8, with the strictly finer subspace refinement in Paper 6), where it carries its bridge conditioning (BA5–BA6, §3). A reader who rejects a bridge axiom loses exactly those readings; the integer and the structural partition proved here are untouched. ¹⁶

Where the cosmological readings live. What this supplement owns is the integer $C_{\text{total}} = 61$, its partition, and the identity I2. The substrate-position-axis architectural reading (the (61, 102) per-slot lattice), the sensitivity table, and the comparison of the bridge readings $\Omega_\Lambda = 42/61$ and kin with the Planck measurements are imported by Papers 8 and 41 at bridge grade (BA5–BA7, §3); that material is staged at those corpus homes and carries the bridge conditioning with it. A reader who rejects a bridge axiom loses exactly those readings and nothing here.

¹⁵Code reference: `check_L_count` in `gauge.py`.

¹⁶Code reference: `check_I2_gauge_cosmological`, `check_T_ACC_unification` in `unification.py`.

A. Comparison with “SM from principles” programs

The comparisons kept in this document are the ones that bear on the classification claim; the reconstruction-program and operational-axiom positioning of the foundational chain lives in Technical Supplement II (*The Foundational Gauge Program*)’s comparison appendix.

A.1 SM-from-principles programs

A separate tradition attempts to derive the Standard Model within an assumed quantum field theory framework.

Anomaly-based selection (Minahan–Ramond–Warner, 1990). Given $SU(3) \times SU(2) \times U(1)$ and the requirement of anomaly cancellation, the SM fermion content is strongly constrained but not unique without additional assumptions (e.g., minimality, family structure). The APF’s T_{field} performs a similar scan, and the gauge template it consumes (input C2 of Theorem 5.9) is not an unexamined assumption: it is the output of the graded APF chain (Theorem R, Technical Supplement II (*The Foundational Gauge Program*)) together with this document’s $L_{\text{gauge,unique}}$ ranking, at the grades stated there.

Grand unification (Georgi–Glashow, 1974). Embeds $SU(3) \times SU(2) \times U(1)$ into $SU(5)$. Explains charge quantization and predicts proton decay. The APF derives the product structure as capacity-minimal (Proposition 4.12: $SU(5)$ costs $2\times$) and obtains the charge-ratio quantization — the unique rational ray of hypercharges — from anomaly cancellation (Theorem 6.1) without grand unification; the absolute charge unit is the compact- $U(1)$ /character-lattice convention, not an anomaly output (§7).

Connes NCG (1996). The noncommutative geometry program derives the SM Lagrangian from a spectral triple. The input is a specific finite-dimensional algebra (chosen to reproduce the SM). The APF obtains the algebra from the foundational chain at its banked grades (T_{alg} , derived given the IJC occupancy; the program document is Technical Supplement II (*The Foundational Gauge Program*)), rather than postulating it; the two programs converge on the same algebraic structure but arrive from opposite directions.

Question	NCG route	APF route
Finite algebra	Chosen as part of the spectral triple data.	Derived from finite admissibility, nonclosure, and Wedderburn decomposition.
Gauge fields	Inner fluctuations / inner automorphisms of the algebra.	Internal continuation-profile-preserving automorphisms, identified with $\text{Inn}(\mathcal{A})$.
U(1) source	Encoded in the chosen finite algebra and unimodularity conditions.	Relative-phase torus of $U(\mathcal{A})$, pruned to one direction by capacity minimality and anomaly compatibility.
Global quotient	Determined by representation of the chosen algebra on the Hilbert space.	Derived from the kernel acting trivially on the scanned matter multiplets.
Anomaly handling	Standard consistency requirement on the represented fermions.	Standard anomaly algebra applied after APF fixes the template and excludes exotic survivors.
Novelty claim	Geometric/spectral encoding of known SM data.	Derivation claim for why this finite algebra/template is the minimum-cost admissible one.

Subtle point	NCG mechanism	APF mechanism
Single U(1)	Unimodularity and central-extension choices restrict the unitary group of the finite spectral triple.	Relative-phase torus is enumerated first, then no-redundancy/A2 and anomaly compatibility select one primitive rational ray.
Hypercharge normalization	Fixed by representation of the finite algebra on the Hilbert space and unimodularity convention.	Anomaly equations determine a ray; primitive integer normalization $Y_{\text{int}} = 6Y$ fixes the charge lattice, while gauge coupling remains conventional.
Global quotient	Kernel of the represented finite-algebra unitary group on the fermion Hilbert space.	Kernel of the lifted $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ action on the scanned matter modules; Proposition 2.26 of Technical Supplement II (<i>The Foundational Gauge Program</i>) verifies it multiplet by multiplet.
Higgs/inner fluctuation	Higgs arises as an inner fluctuation of the finite Dirac operator.	Higgs appears as a capacity-minimal scalar bridge for Yukawa/admissible mass terms; its dynamical potential is downstream, not used to derive the gauge template here.
Finite algebra input	Chosen as spectral-triple data subject to NCG axioms.	Produced as the minimum-cost finite admissibility carrier, with NCG-like algebraic facts applied after selection.

Quaternionic branch and weak SU(2). NCG often realizes the weak factor through an almost-commutative algebra containing an $\mathbb{H}$ summand. APF does not use that route. The weak factor appears as the simply connected lift of the PU(2) internal automorphism factor acting on complex doublet matter modules. Quaternionic simple components are not part of the Paper 2 finite complex-sector theorem: within the imported Paper 5 field-selection gate, they either collapse the chirality/charged-module distinction needed for the scanned template or add an anti-linear/reality structure not selected by the no-added-record criterion. Thus APF replaces the quaternionic ingredi-

ent by complex doublet modules plus the cover $\mathrm{PU}(2) \leftarrow \mathrm{SU}(2)$; the full real/complex/quaternionic exclusion is imported from the field-selection supplement.

Lattice chiral-gauge and symmetric-mass-generation routes. Lattice chiral-gauge programs, mirror-fermion decoupling, symmetric mass generation, and related RKD/disentangler constructions address a different problem: whether a given anomaly-free chiral spectrum can be realized by a local microscopic regulator without unwanted light mirror sectors. APF uses anomaly cancellation as a necessary filter, not as a sufficient proof of lattice realizability. The finite-capacity derivation therefore should be read as an upstream selection theorem for the candidate low-energy spectrum; a lattice construction would be an independent downstream regulator witness. If an anomaly-free APF survivor failed all local-regulator tests, that would be evidence against its physical admissibility, not a contradiction of the anomaly algebra.

In the other direction, SMG/mirror-fermion successes support the APF split between algebraic consistency and microscopic realizability: anomaly-free spectra can sometimes be gapped without breaking symmetry, while anomalous spectra cannot. APF’s additional claim is that the minimal finite-capacity ledger selects the SM survivor before choosing a regulator. The two questions are therefore complementary: APF asks which template is minimum-cost; lattice work asks whether the template admits a local UV completion.

Doplicher–Roberts reconstruction and Pinzari’s braided extensions. The classical DR theorem (1989) reconstructs a compact group G from its representation category (a symmetric C^* -tensor category with conjugates). In the APF, T3 (Paper 1 Supplement) uses DR to recover the gauge group from the admissibility algebra’s representation category; the gauge identification theorem (Theorem 2.11 of Technical Supplement II (*The Foundational Gauge Program*)) then identifies which automorphisms constitute the gauge group.

Proposition A.1 (Finite APF admissibility net as a DR precursor). *A finite APF admissibility net determines the finite precursor of a Doplicher–Roberts reconstruction datum. The assignment sends an interface Γ to its field algebra $\mathcal{A}(\Gamma)$, sends an inclusion of interfaces to the corresponding isotony inclusion, and sends each admissible charged sector to a finite-dimensional endomorphism/module category of $\mathcal{A}(\Gamma)$. In this finite regime the following DR hypotheses are present: a rigid semisimple tensor category of sectors, conjugates for charged modules, a symmetric exchange rule for independent spacelike interfaces, and a compact internal automorphism group reconstructed as $\mathrm{Inn}(\mathcal{A})$.*

Proof sketch and hypothesis audit. Semisimplicity and finite rigidity follow from the Wedderburn decomposition of $\mathcal{A}(\Gamma)$. Tensor product is the APF composition of independent interfaces; conjugates are provided by dual modules. The local-support lemma gives symmetric interchange for independent finite interfaces, while the continuum/type-III section records the extra hypotheses needed to pass to a Haag–Kastler net: nuclearity/split estimates, Haag duality in the scaling limit, and convergence of finite local algebras to type-III local algebras. Thus the finite APF result is not claimed to be a full DHR theorem in the continuum. It is a finite, cost-equipped precursor whose robust output is the compact internal group and its representation category. The assumptions that remain external to the finite theorem are exactly the continuum analytic hypotheses, not the finite inner-automorphism calculation. $\square$

DR/DHR datum	APF finite counterpart	Continuum status
Local observable/field net	Cost-equipped admissibility field algebra $\mathcal{A}(\Gamma)$ and invariant subalgebra $\mathcal{A}(\Gamma)^G$	Requires scaling limit and Haag–Kastler regularity.
Selection sectors	Admissible charged modules/continuation sectors	Robust as representation category; analytic localization is continuum-dependent.
Tensor product	Independent-interface composition	Finite version exact; continuum version requires split/nuclearity hypotheses.
Compact gauge group	$\text{Inn}(\mathcal{A}) = \prod_k \text{PU}(n_k)$ plus central grading	Finite output; global topology refined by matter-module kernel.
Duality/reconstruction	T3 plus Theorem 2.11 of Technical Supplement II (<i>The Foundational Gauge Program</i>)	Full DHR theorem imported only in the continuum bridge.

Pinzari’s program (2017–) extends DR to *braided* tensor categories and weak Hopf C^* -algebras. In this setting, abelian factors can emerge from the *center* of the fusion category: central objects (transparent to braiding) generate grading tori that act as abelian gauge factors. This is structurally parallel to the APF’s mechanism for $U(1)_Y$: the relative-phase torus $T_{\text{rel}} = Z(U(\mathcal{A}))/U(1)_{\text{diag}}$ (Remark 2.18 of Technical Supplement II (*The Foundational Gauge Program*)) plays the role of Pinzari’s grading torus, and the abelian factor arises from central (= transparent) elements rather than from inner automorphisms.

The key structural parallel is: *non-abelian gauge from inner automorphisms, abelian gauge from the center*. In DR/Pinzari, this is “gauge from the Tannakian group, grading from the Müger center.” In APF, it is “gauge from $\text{Inn}(\mathcal{A})$, grading from T_{rel} pruned by CM.” The APF adds a *selection mechanism* (capacity minimality, Proposition 7.1) that is absent in the DR/Pinzari framework: the latter reconstructs whatever symmetry the category has, while the APF derives the *minimal* symmetry compatible with admissibility completeness.

Hopf, weak-Hopf, VOA, and quantum-group reconstructions. Weak-Hopf, quantum-group, VOA, and categorical reconstruction programs allow more general “quantum symmetry” than ordinary compact Lie-group inner automorphisms. APF intentionally restricts the finite Standard-Model core to ordinary inner automorphisms of a semisimple admissibility algebra because the objects being represented are classical-stable record distinctions at a finite interface. More general quantum symmetries would require additional braided or fusion data in the carrier. Such extensions are not forbidden by APF, but they are not part of the minimum finite-interface theorem; they would be higher-structure additions whose admissibility cost must be justified separately.

Finite tensor categories and symmetric tensor categories. Recent work on finite tensor categories (Etingof–Gelaki–Nikshych–Ostrik) and symmetric tensor categories (Deligne’s theorem and extensions by Coulembier, Entova-Aizenbud, Ostrik) shows that structural constraints on representation categories can have sharp classification consequences. Deligne’s theorem, for example, classifies all symmetric tensor categories of moderate growth as representation categories of supergroups — a result that constrains the possible internal symmetry structures of any theory whose observables form a symmetric tensor category.

The APF is not a tensor-categorical framework: its primary objects are admissibility algebras and cost functionals, not abstract tensor categories. However, the representation category of the admissibility algebra $\mathcal{A} \cong \bigoplus_k M_{n_k}(\mathbb{C})$ is a finite semisimple tensor category (the category of

finite-dimensional $\mathcal{A}$ -modules), and its symmetry is classified by the theorems above. The APF’s contribution relative to this literature is not a new classification of tensor categories but a *physical derivation* of which tensor category is realized: the admissibility postulate A1 selects $\mathcal{A}$ and hence its representation category, rather than postulating the category directly. The classification theorems then apply as *imported mathematics* (same epistemic status as Wedderburn–Artin or Skolem–Noether in the present supplement).

Relation to L_∞/C_∞ and homotopy-algebraic gauge formalisms. Homotopy-algebraic approaches encode the consistency identities of gauge theories—gauge closure, higher brackets, deformation theory, and the organization of interactions—once the field complex and symmetry data are specified. The APF is aimed one step upstream. The finite admissibility algebra itself is not asserted to carry a primitive L_∞ or C_∞ structure. Rather, once APF has fixed the field complex and gauge template, the usual BV/BRST or homotopy-algebraic lift may be constructed to organize interactions and gauge identities. In this sense the two programs are complementary: homotopy algebras organize consistent dynamics, while APF attempts to derive the minimum-cost admissible input data. **Topological charge quantization and $\mathbb{CP}^k$ sigma-model routes.** Topological constructions can quantize charges by assigning matter to bundles or soliton sectors and imposing integrality of topological classes. APF’s hypercharge quantization is different in scope. It does not use a background $\mathbb{CP}^k$ target space or a topological winding number; it first fixes the finite admissibility algebra and chiral template, then applies the standard anomaly equations to obtain a unique rational ray. Consequently APF’s additional claim is not merely charge quantization, but exclusion of redundant abelian directions within the minimal finite template. If a topological route introduces extra sectors or sterile records, APF classifies that as an enlarged admissibility problem rather than a counterexample to the minimal theorem.

EFT anomaly routing and Goldstone contributions. Effective-field-theory analyses remind us that anomaly terms can be redistributed among current definitions, Wess–Zumino terms, and Goldstone contributions. The present supplement uses anomaly cancellation only at the renormalizable chiral-template level: F6 checks the UV gauge consistency of the candidate fermion spectrum before EFT operators or symmetry-breaking field redefinitions are introduced. APF therefore does not infer new low-energy anomaly coefficients from the ledger; it uses the standard exact anomaly equations as a consistency filter on the APF-selected template.

B. Code–theorem concordance

Every theorem in this supplement — and every chain theorem it consumes from Technical Supplement II (*The Foundational Gauge Program*) — is verified by a named check function in the codebase. The table below provides the mapping; rows whose section column points into Technical Supplement II (*The Foundational Gauge Program*) are consumed inputs, banked under the same verification contract.

Theorem	Supplement §	Function	File
L_{Cauchy}	§1 of Technical Supplement II (<i>The Foundational Gauge Program</i>)	check_L_Cauchy_uniqueness	L_Cauchy_uniqueness.py
Gauge ident.	§2.3 of Technical Supplement II (<i>The Foundational Gauge Program</i>)	check_T3	core.py
T_{7B}	§3.1 of Technical Supplement II (<i>The Foundational Gauge Program</i>)	check_T7B	gravity.py
$L_{\text{irr,uniform}}$	§3.2 of Technical Supplement II (<i>The Foundational Gauge Program</i>)	check_L_irr_uniform	core.py
$B1'$	§3.3 of Technical Supplement II (<i>The Foundational Gauge Program</i>)	check_B1_prime	gauge.py
$L_{\text{gauge,unique}}$	§4	check_L_gauge_template_uniqueness	gauge.py
T_{field}	§5	check_T_field; fermion_scan_standalone.py [20]	gauge.py
$L_{\text{hypercharge}}$	§6	check_T_field (hypercharges)	gauge.py
L_{count}	§10	check_L_count	gauge.py
$I2$ (gauge-cosmo)	§10.3	check_I2_gauge_cosmological, check_T_ACC_unification	unification.py
P_{exhaust}	exited (Paper 8; §3)	check_P_exhaust	core.py
L_{equip}	exited (Paper 41; §3)	check_L_equip	cosmology.py
$L_{\text{sat,part}}$	exited (Paper 8; §3)	check_L_saturation_partition	cosmology.py
Δ_{sig}	exited (Paper 6; §3)	check_Delta_signature	spacetime.py
T_8 ($d=4$)	exited (Paper 6; §3)	check_T8	spacetime.py
$T_{9,\text{grav}}$	exited (Paper 6; §3)	check_T9_grav	gravity.py
$T_{\theta\text{QCD}}$	exited (Paper 46; §3)	check_T_theta_QCD	gauge.py
$T_{\text{vac,stab}}$	exited (Paper 45; §3)	check_T_vacuum_stability	gauge.py
T_{proton}	exited (bank; BA8 in §3)	check_T_proton	gauge.py

The full codebase is available at github.com/Ethan-Brooke/APF-Paper-2-The-Structure-of-Admissible-Physics.

Reproducibility of the fermion scan. The function `check_T_field` in `apf/gauge.py` enumerates the template space in ordered-tuple form (4,680 iterations over the 1,680 distinct templates) and applies the seven filters under the uniform-dimension F6 variant — the 4/2 robustness path of the predicate taxonomy (§5.3) — verifying that the unique minimum-DOF survivor is the SM at 45 Weyl fermions. It records no per-stage waterfall counts. The canonical full-system waterfall (10/5; every count in §5.4) is generated by `check_L_F6_not_from_EC` in `apf/ec_inventory_reading.py` and by the release’s audit files. The exclusion-proof assertions (P1–P5) are verified within `check_T_field`; hypercharge derivation and anomaly cancellation are checked algebraically in exact rational arithmetic (`fractions.Fraction`). Running `python verify_all.py` from the repository root runs the full check registry (both functions included) with no dependencies beyond the Python standard library.

Standalone verification script. The file `fermion_scan_standalone.py` [20], included in the GitHub repository and archived on Zenodo (<https://doi.org/10.5281/zenodo.19154197>), is a self-contained Python script (~550 lines, zero dependencies beyond the standard library) that reproduces the complete scan from scratch. An interactive Colab walkthrough with waterfall visualisation is available at https://colab.research.google.com/drive/1ot01Qwp_Lr30sK6ykkeMRATanecPkKg6. Running `python3 fermion_scan_standalone.py` performs the following:

- (1) Enumerates all 1,680 distinct templates using `combinations_with_replacement`.
- (2) Applies all 7 filters with exact rational arithmetic, including spectator-field reduction for $A(R) = 0$ representations.
- (3) Prints the waterfall table with exact counts at each stage.
- (4) Lists all 8 post-F6 survivors under its spectator-reduction predicate (the robustness form; the canonical full-system listing in §5.4 carries 10).
- (5) Lists all 4 post-CPT survivors (spectator-reduction form; the canonical count is 5).
- (6) Verifies all 5 exclusion proofs (P1–P5) with explicit arithmetic.
- (7) Derives both hypercharge solutions ($z = 1 \pm N_c$), verifies all 4 anomaly conditions exactly for each, and confirms their physical equivalence under $u \leftrightarrow d$ relabelling.
- (8) Runs 4 built-in red-team challenges: (RT1) combinatorial cross-check of the enumeration count, (RT2) explicit spectator-field anomaly verification, (RT3) dead-code reachability audit, (RT4) the two-colored-doublet class-dominance enumeration (class minimum 54 Weyl DOF, witness-verified; Proposition 5.5).
- (9) Asserts 12 final verification conditions.

The script is designed so that any reader can verify the entire fermion-content derivation on a standard laptop in under one second, with no knowledge of the APF codebase required.

The Lie-algebra classification (§4) is verified by `check_L_gauge_template_uniqueness`, which tests the classification’s 17 representatives (the families closed by the monotonicity arguments of §4) against R1–R3 and records the exclusion reason for each. The Zenodo archive (DOI: 10.5281/zenodo.18867119) provides a frozen snapshot of the codebase corresponding to this supplement.

C. Fermion scan: algorithmic walkthrough

This appendix makes the fermion-content derivation (§5) verifiable without running code. It specifies the search space construction, gives the algorithm in pseudocode, and traces the critical filter steps on explicit examples.

C.1 Search space construction

A template is a multiset of $(r^{(3)}, r^{(2)})$ pairs. The search space is bounded by P1–P5 (§5.2):

Allowed SU(3) reps (P1: $\dim < 10$): $\{3, \bar{3}, 6, \bar{6}, 8\}$. Dynkin indices: $T(3) = T(\bar{3}) = 1/2$, $T(6) = T(\bar{6}) = 5/2$, $T(8) = 3$. Cubic anomaly coefficients: $A(3) = 1/2$, $A(\bar{3}) = -1/2$, $A(6) = 7/2$, $A(\bar{6}) = -7/2$, $A(8) = 0$. (The quadratic *index* $T(6) = 5/2$ and the cubic *anomaly* $A(6) = 7/2$ are distinct quantities; v3.9’s appendix carried 5/2 for both, a holdover from before the v3.7 correction. The value is computationally inert for the waterfall: sextet-containing templates do reach F4, but no F4 verdict differs between $A(6) = 5/2$ and $7/2$ anywhere in the reachable space — verified by direct recomputation of all 144 sextet templates passing F1–F3, 2026-07-13. The v3.7 changelog’s stated reason, that sextets are AF-eliminated before F4, was incorrect.)

Allowed SU(2) reps (P2 for colored fields; P3 dominance for colorless triplets): $\{1, 2\}$. Dynkin indices: $T(1) = 0$, $T(2) = 1/2$.

Template structure (P4–P5): Each template contains exactly one colored doublet (P4: two colored doublets force ≥ 54 DOF, class dominance), 0–3 colored singlets, 0–1 lepton doublets, and 0–2 lepton singlets (P5: extra field types beyond the anomaly-closed core cannot lower the minimum — dominance).

Enumeration count. The colored doublet ranges over 5 reps. The colored singlets form a multiset of size 0–3 drawn from 5 reps: $\binom{5+0-1}{0} + \binom{5+1-1}{1} + \binom{5+2-1}{2} + \binom{5+3-1}{3} = 1 + 5 + 15 + 35 = 56$ multisets. The lepton doublet is present or absent: 2 choices. The lepton singlets: 0, 1, or 2 copies of $(\mathbf{1}, \mathbf{1})$: 3 choices. Total: $5 \times 56 \times 2 \times 3 = 1,680$ distinct templates. The standalone verification script [20] uses `combinations_with_replacement` to enumerate these directly, and verifies the count against this closed-form formula as a built-in red-team challenge (RT1).

C.2 Algorithm in pseudocode

```
INPUT:  SU3_reps = {3, 3b, 6, 6b, 8}
        SU2_reps = {1, 2}
        N_gen = 3
        AF3_bound = 11, AF2_bound = 22/3
        AF_coeff = 2/3
```

```

ENUMERATE all templates T:
  FOR each colored_doublet IN SU3_reps:           // 5
    FOR n_colored_singlets IN {0, 1, 2, 3}:
      FOR each combo IN C(SU3_reps, n_cs):        // combinations w/ replacement
        FOR has_lepton_doublet IN {True, False}:  // 2
          FOR n_lepton_singlets IN {0, 1, 2}:      // 3
            T = [(colored_doublet, 2)]
                + [(c, 1) for c in combo]
                + [(1, 2)] if has_lepton_doublet
                + [(1, 1)] * n_lepton_singlets
            total_tested += 1

            // F1+F2: Asymptotic freedom (exact rational)
            su3_cost = SUM over (a,b) in T:
                T_3(a) * dim_2(b) * N_gen
            su2_cost = SUM over (a,b) in T:
                T_2(b) * dim_3(a) * N_gen
            IF AF3_bound - AF_coeff * su3_cost <= 0: SKIP
            IF AF2_bound - AF_coeff * su2_cost <= 0: SKIP

            // F3: Doublet-singlet content
            IF no colored doublet or no colored singlet: SKIP

            // F4: [SU(3)]^3 anomaly cancellation
            IF SUM A_3(a) * dim_2(b) != 0: SKIP

            // F5: Witten anomaly
            IF SUM dim_3(a) [where b=2] is odd: SKIP

            // F6: Full anomaly system
            // (rational root test on the hypercharge
            // quadratic -- details in §8)
            IF no rational hypercharge solution: SKIP

            // F7: CPT quotient
            canonical = min(sorted(T),
                           sorted(conjugate(T)))
            IF canonical already seen: SKIP
            Mark canonical as seen

            // Record survivor
            DOF = SUM dim_3(a) * dim_2(b) * N_gen
            survivors.append((DOF, T))

SELECT minimum-DOF survivor(s)

```

C.3 Filter-by-filter trace on critical examples

We trace three templates through all seven filters: the SM (passes all), a near-miss (fails at F6), and a common failure (fails at F4).

Example 1: SM template $\{(3, 2), (\bar{3}, 1), (\bar{3}, 1), (1, 2), (1, 1)\}$.

$$F1 (SU(3) AF): \quad b_0^{(3)} = 11 - \frac{2}{3} \left[\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 \right] \cdot 3 = 11 - \frac{2}{3} \cdot 2 \cdot 3 = 11 - 4 = 7 > 0. \quad \checkmark$$

F2 (SU(2) AF): $b_0^{(2)} = \frac{22}{3} - \frac{2}{3}[\frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 1] \cdot 3 = \frac{22}{3} - \frac{2}{3} \cdot 2 \cdot 3 = \frac{22}{3} - 4 = \frac{10}{3} > 0$. ✓

F3 (doublet–singlet content): $(3, 2)$ is a colored doublet; $(\bar{3}, 1)$ is a colored singlet. Both present. ✓

F4 ([SU(3)]³): $\frac{1}{2} \cdot 2 + (-\frac{1}{2}) \cdot 1 + (-\frac{1}{2}) \cdot 1 = 1 - \frac{1}{2} - \frac{1}{2} = 0$. ✓

F5 (Witten): Doublets weighted by SU(3) dim: $3 + 1 = 4$ (even). ✓

F6 (full anomaly): $N_c = 3$, $n_{cs} = 2$, $n_{ls} = 1$. Discriminant $= 4 + 4(9 - 1) = 36$. $\sqrt{36} = 6 \in \mathbb{Z}$. ✓

F7 (CPT): Canonical form $= \min(\text{sorted}(T), \text{sorted}(\bar{T}))$. $\bar{T} = \{(\bar{3}, 2), (3, 1), (3, 1), (1, 2), (1, 1)\}$. Sorted: $(1, 1)(1, 2)(3, 2)(\bar{3}, 1)(\bar{3}, 1)$ vs. $(1, 1)(1, 2)(3, 1)(3, 1)(\bar{3}, 2)$. First is lexicographically smaller. Canonical = SM. ✓

DOF $= (6 + 3 + 3 + 2 + 1) \times 3 = 45$. **Passes all filters.**

Example 2: $\{(6, 2), (\bar{6}, 1), (1, 2), (1, 1)\}$ (a template with the sextet).

F1: $b_0^{(3)} = 11 - \frac{2}{3}[\frac{5}{2} \cdot 2 + \frac{5}{2} \cdot 1] \cdot 3 = 11 - \frac{2}{3} \cdot \frac{15}{2} \cdot 3 = 11 - 15 < 0$. × **Fails F1.**

Example 3: $\{(3, 2), (3, 1), (1, 2), (1, 1)\}$ (wrong-sign colored singlet).

F1–F3: Pass (verify: $b_0^{(3)} = 11 - \frac{2}{3} \cdot \frac{3}{2} \cdot 3 = 8 > 0$; both content types present).

F4: $\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 1 = \frac{3}{2} \neq 0$. × **Fails F4.** The cubic anomaly is not cancelled because both colored fields are in the **3** (same sign of A).

Example 4: $\{(3, 3)_0, (\bar{3}, 1)_{+1}, (\bar{3}, 1)_{-1}, (\bar{3}, 1)_0\}$ (the colored weak-triplet content — the sharpest known rival, 18 Weyl fermions as a single copy).

This content is chiral, satisfies every gauge and gravitational anomaly condition exactly, satisfies Witten parity (no doublets), and as a *single copy* is asymptotically free in both non-abelian factors — it is the strongest stress test of the scan’s completeness, and it is excluded at every multiplicity by three independent banked routes. At $n = 1$ or 2 it is an under-replicated rival, killed by the max-saturated selection¹⁷ and by the capacity–beta identities, which no $n \neq 3$ satisfies¹⁸. At $n \geq 2$ it destroys SU(2) asymptotic freedom outright $((2/3)(2)(3)(2) = 8 > 22/3$; at the forced $n = 3$, $12 > 22/3$ — Proposition P2’s arithmetic). And its own capacity–beta system has *no* solution at any n : the non-abelian identities give $n = 27/13$ and $n = 23/10$, non-integer, with the convention-free $6|b_3|$ equation failing every n by a margin of at least 18. The Standard-Model template satisfies all three identities simultaneously at $n = 3$; this rival satisfies none at any n .

C.4 Completeness argument

The scan’s boundedness rests on three facts:

1. **Representation bound (P1–P2, theorems).** Every SU(3) rep with $\dim \geq 10$ and every colored SU(2) rep with $\dim \geq 3$ exceeds the AF budget as a *single field* (the cost of one such field, summed over 3 generations, exceeds b_0). These are closed-form inequalities verified in §5.2. Therefore the allowed reps are exactly $\text{SU}(3) \in \{3, \bar{3}, 6, \bar{6}, 8\}$ and $\text{SU}(2) \in \{1, 2\}$.

¹⁷Code reference: `check_R_Ngen_neq_3_killed` in `killed_rivals.py`.

¹⁸Code reference: `check_L_beta_capacity` in `generations.py`.

2. **Multiplicity caps (declared inputs; P3–P5 dominance).** At most 1 colored doublet (P4: two colored doublets force ≥ 54 DOF — the class-dominance enumeration of Proposition 5.5), at most 3 colored singlets, at most one lepton doublet, at most 2 lepton singlets — template size $|\mathcal{T}| \leq 7$. These caps are declared enumeration inputs (§5.2). P3 and P5 are dominance results: within the declared space, every survivor carrying types beyond the five SM roles extends an anomaly-closed core and sits strictly above it in DOF, so the enumerated enlargement directions cannot lower the minimum.
3. **Complete enumeration within the caps.** Within the bounded set ($5 \times 56 \times 2 \times 3 = 1,680$ distinct templates), every combinatorial possibility is generated and tested. No branch is skipped. The enumeration is deterministic and order-independent (the final result depends only on the filter outcomes, not on the order of testing).

Together, (1)–(3) establish the licensed claim: every chiral fermion template within the declared representation bounds and multiplicity caps is either in the 1,680-template set or is excluded by a closed-form proof, and the SM at 45 Weyl DOF is the unique minimum *among the enumerated candidates*. Completeness of the caps themselves — that no admissible sub-45-DOF template exists outside them — is the open proof obligation of the reproducibility contract (§5), carried by P3/P5-style dominance for the enumerated directions and otherwise open.

C.5 Robustness checks

1. **Exact arithmetic.** All filter computations use Python’s `fractions.Fraction` (exact rational arithmetic). No floating-point approximation is made at any stage. The anomaly conditions are checked as exact equalities, not as $|\text{residual}| < \epsilon$.
2. **Filter-order independence.** The final survivor set is determined by the conjunction of all seven filters. The sequential application (F1 before F2 before F3, etc.) is an optimization (early termination), not a logical dependency. Reordering the filters produces the same final set. The standalone script (`fermion_scan_standalone.py`) applies all filters and records intermediate counts; permuting the filter order and comparing outputs verifies independence.
3. **CPT symmetry.** The CPT quotient (F7) identifies each template with its charge conjugate. The conjugation map $3 \leftrightarrow \bar{3}$, $6 \leftrightarrow \bar{6}$, $8 \rightarrow 8$, $1 \rightarrow 1$ is an involution. The canonical representative is the lexicographic minimum of the pair $\{T, \bar{T}\}$. The 10 post-F6 survivors under the canonical full-system predicate form 5 CPT pairs (listed in §5.4); the standalone script’s spectator-reduction predicate yields 8, forming 4 pairs.
4. **Sensitivity to generation count.** The scan assumes $N_{\text{gen}} = 3$ (derived in Paper 2, §9). With $N_{\text{gen}} = 1$ or 2: the AF bounds loosen (b_0 terms scale as N_{gen}), admitting additional survivors. With $N_{\text{gen}} \geq 4$: the AF bounds tighten, reducing or eliminating survivors. The $N_{\text{gen}} = 3$ scan is therefore not vacuous (it admits templates) and not trivial (it excludes most). The looser $N_{\text{gen}} \in \{1, 2\}$ survivor sets are not an open door: those generation counts are themselves excluded (the capacity–beta identities and the rival kill, §8), so their would-be survivors are killed at the multiplicity level before content filtering arises.
5. **Second survivor.** The second CPT-inequivalent survivor (DOF = 48, SM + extra singlet) provides a falsifiability test: if the SM contained 48 Weyl fermions (e.g., with a right-handed neutrino), the scan’s minimum would move off the Standard-Model template, and the count would move to $C = 64$ — whose bridge-conditional comparison with observation (BA6) lives at the corpus homes (§9).

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